Degree Final Project

Degree in Industrial Tecnologies Engineering (GETI)

Model Based Torque Vectoring for a 4WD Formula Student Vehicle

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Abstract

The objetive of this project is to learn, design, implement, and analyze the results of Torque Vectoring for a Formula Student racing vehicle using a model based controller. The entire process of this controller can be achieved through various elements. To begin with, acquiring theoretical knowledge in dynamics, signals, optimization functions, among others. From this point, implementation starts, encountering problems and finding solutions until achieving the desired controller.

Another goal, and the most important one, is to implement Torque Vectoring for the BCNeMotorsport vehicle for the upcoming 2023-2024 season with the CAT16x, aiming to reach a new level of performance surpassing previous years and demonstrating it in the summer competitions.

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Nomenclature

Vehicle Variables

C_d Drag coefficient [-]

C_l Lift coefficient [-]

CoG Vehicle Center of Gravity $[kg\ m^2]$

h_{CoG} Vehicle Center of Vehicule Height $[m]$

i Transmission Ratio $[-]$

J_v Vehicle Yaw Inertia $[kg\ m^2]$

J_w Wheel Pitch Inertia $[kg\ m^2]$

l Vehicle Wheelbase $[m]$

l_f Vehicle distance from CoG to front axle $[m]$

l_r Vehicle distance from CoG to rear axle $[m]$

m Vehicle mass $[kg]$

m_i Vehicle mass of the track i $[kg]$

R_w Wheel Radius $[kg\ m^2]$

t_f Vehicle front track width $[m]$

t_r Vehicle rear track width $[m]$

Kinematics and Dynamics Variables

α_F Slip angle of the front axle $[rad]$

α_i Slip angle for i tire $[rad]$

α_R Slip angle of the rear axle $[rad]$

β Vehicle Slip angle at the CoG $[rad]$

$\ddot{\psi}$ Vehicle Yaw Acceleration $[rad/s^2]$

ΔW_{lat} Lateral Load transfered $[N]$

ΔW_{lon} Longitudinal Load transfered $[N]$

δ Driver Steering Angle $[rad]$

δ_w Wheel Angle $[rad]$

$\dot{\psi}$ Vehicle Yaw Rate $[rad/s]$

$\dot{w}_w$ Wheel angular acceleration $[rad/s^2]$

σ Slip ratio $[-]$

a_x Vehicle Longitudinal Acceleration $[m/s^2]$

a_y Vehicle Lateral Acceleration $[m/s^2]$

C_f Cornering stiffness of the front axle $[N/rad]$

C_r Cornering stiffness of the rear axle $[N/rad]$

$F_{x,i}$ Longitudinal force for i tire $[N]$

$F_{y,i}$ Lateral force for i tire $[N]$

g Gravity $[m/s^2]$

M_z Total Yaw moment of the vehicle $[Nm]$

$M_{z,Fy}$ Yaw moment of the vehicle due to lateral forces $[Nm]$

v_x Vehicle Longitudinal Velocity $[m/s]$

v_y Vehicle Lateral Velocity $[m/s]$

$v_{x,i}$ Longitudinal velocity for i tire $[m/s]$

$v_{y,i}$ Lateral velocity for i tire $[m/s]$

w_w Wheel angular velocity $[rad/s]$

$W_{s,i}$ Static Load for i tire $[N]$

Control Variables

α_n Priority coefficients for each of the TD objectives $[-]$

Γ_i Commanded torque to i wheel $[Nm]$

F_x^{ref} Desired value for the F_x in the TD problem $[N]$

$F_{x,max}$ Upper limit for the F_x variable in the TD problem $[N]$

$F_{x,min}$ Lower limit for the F_x variable in the TD problem $[N]$

I_b Intensity of the battery $[A]$

J_n Objectives of the TD problem $[-]$

K_i Integral variable for the PI controller $[Nm/rad]$

K_p Proportional variable for the PI controller $[Nm/rad]$

M_z^{ref} Desired value for the M_z in the TD problem $[Nm]$

$M_{z,max}$ Upper limit for the M_z variable in the TD problem $[Nm]$

$M_{z,min}$ Lower limit for the M_z variable in the TD problem $[Nm]$

$M_{z,TV}$ Yaw moment due to the Torque Vectoring $[Nm]$

P_{max} Upper limit for the developed power in the TD problem $[W]$

t_s Simulation time $[s]$

V_b Battery voltage $[V]$

Acronyms

PU Processing Unit

GPS Global Positioning System

IMU Inertial Measurement Unit

INS Inertial Navigator System

QP Quadratic Problems

PID Proportional-Integral-Derivative

FF Feed-Forward

FB Feed-Back

TD Torque Distribution

TA Torque Allocator

TV Torque Vectoring

TC Traction Control

PC Power Control

ROS Robotic Operating System

CAN Controller Area Network

LTV Linear Time Varying

NTV Non Linear Time Varying

TPDO Transmit Process Data Object

RPDO Receive Process Data Object

EV Electric Vehicles

DV Driverless Vehicles

CV Combustion Vehicles

DC Driverless Cup

DQ Disqualification

DOF Degrees of Freedom

1 Introduction

1.1 Motivation

In the automotive industry, technological innovation has become the driving force behind the quest for performance and efficiency. Formula Student is a globally recognized competition that challenges engineering students to design, build, and race their own single-seater racing cars. This exciting platform not only demands technical skills but also encourages creative thinking in vehicle dynamics and control systems. It is in this lively and competitive environment that the motivation for this thesis originates.

At the center of this effort lies the BCNeMotorsport team, a group of engineering students who form the core of this ambitious project. Among the team's departments, the Vehicle Controls division distinguishes itself with its distinguished capabilities and innovative techniques. This department focuses on creating and implementing advanced control systems to improve the performance and handling of their Formula Student vehicle.

This thesis embarks on an exploration of Torque Vectoring, an advanced technology that is reshaping vehicle dynamics. By adjusting the torque delivered to individual wheels, Torque Vectoring has the potential to significantly enhance the performance of a Formula Student car. The algorithm distributes torque between the four wheels in real-time, improving traction, stability, and cornering capabilities. Following an exhaustive examination, this thesis aims to uncover the potential advantages and challenges of integrating Torque Vectoring into a Formula Student car. Ultimately, it seeks to push the boundaries of what is achievable in this highly competitive field.

1.2 State of the art

The current state of the automotive industry is marked by a significant shift towards hybrid and electric mobility, showcasing a remarkable surge in innovation. The increasing popularity of electric vehicles (EVs) comes from their substantial advantages, including reduced emissions and lower operating costs. This transition has paved the way for innovative advances in technologies like Torque Vectoring (TV) algorithms. By integrating TV in electric vehicles, engineers can benefit the precise control of power distribution to enhance traction and stability, ultimately optimizing the overall driving experience. This innovation aligns with the broader goal of achieving sustainable and efficient transportation solutions, reflecting the evolving landscape of modern automotive engineering.

Other companies, such as Porsche, have opted for an electromechanical power steering system [2], which provides more accurate control, lower wear, and high efficiency. Other methods used include the implementation of rear-wheel steering, which is much more complex to design but offers certain advantages in controlling the vehicle's behavior [3].



Figure 1.1: Porsche Cayman GT4 RS.

Regarding the yaw rate controller, common approaches have been studied, including the use of a Proportional-Integral-Derivative (PID). Alongside controller design, it is essential to develop a control-oriented model. A fairly common approach involves the use of a bicycle model that depends on speed and wheel steering, a method that some studies have improved by including a segmented tire model. Other approaches have been made, such as a control method for front-to-rear torque vectoring distributions, as described in [6]. Further improvements in control can be seen in [1], where the use of stability yaw rate techniques or sideslip correction demonstrates significant enhancement in certain cases.

As for the Torque Distribution (TD) issue, there are common approaches that base power distribution on each side of the vehicle (left to right). While this may be suitable in the case of vehicles with two power-train actuators, it requires further distribution over the front and rear wheels on each side, based on vertical loads, especially when motors are also integrated into the front axle, without overlooking slip angles. Apart from this and continuing with the work of the team from previous years, for the bases of the TA [13] has been used.

1.3 Objectives

The main goal of this thesis is to design and implement a Torque Vectoring control system for a Formula Student single-seater. Its powertrain is distributed across four motors, one on each wheel, independent of each other. This primary objective can be broken down into the following:

- Present all the dynamic theory behind this type of controller, which varies the behaviour and performance of the vevhicle. This includes the dynamic aspect of the vehicle, as well as its components and approximations of different models to represent it.

- Define a model-oriented control. The chosen model is a dual track bicycle model, calibrated to the CAT15x using data obtained during the testing season. This model has the characteristic of being able to manipulate the behavior of the vehicle allowing the adjustment of many more parameters to vary its behavior compared to a linear bicycle model.

- Design a yaw rate controller to generate the reference yaw moment, which is the most important component of the algorithm for achieving the desired vehicle behavior. This controller is highly functional as it can be adjusted depending on the desired behavior in real-time.

- Define and address the different objectives of the multi-objective optimization function. Both physical constrains and those desired by the user must be taken into account, as well as the goals of the function.

- In the design phase, implement the entire control system in Simulink for use with the VI-Grade CarRealTime simulation software tool, this involves defining the entire vehicle model. Also implementing it on the Processing Unit (PU) of the CAT15x is required in the Robotic Operating System (ROS) to ensure proper communication with the vehicle and compatibility with other algorithms.

- Analyze the collected data and behavior of the single-seater, both in simulation and in the real prototype, to verify the different references and the robustness of the algorithm in order to achieve maximum performance.

1.4 Project Outline

In this section, there is an outline in the figure 1.2 shown to help understand how this project is set up. Along with this picture, an introduction to all the mentioned chapters is given.

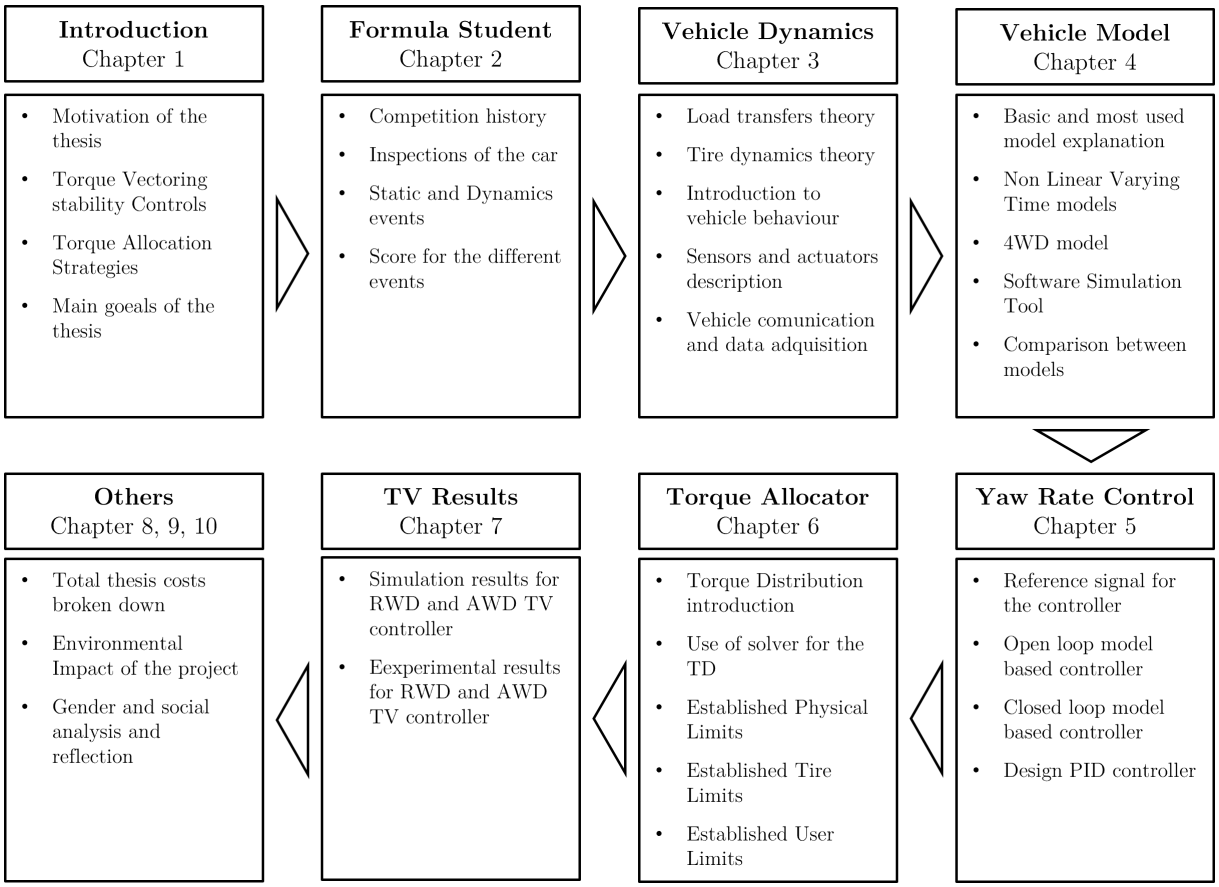


Figure 1.2: Project Outline.

1.5 Gantt Diagram

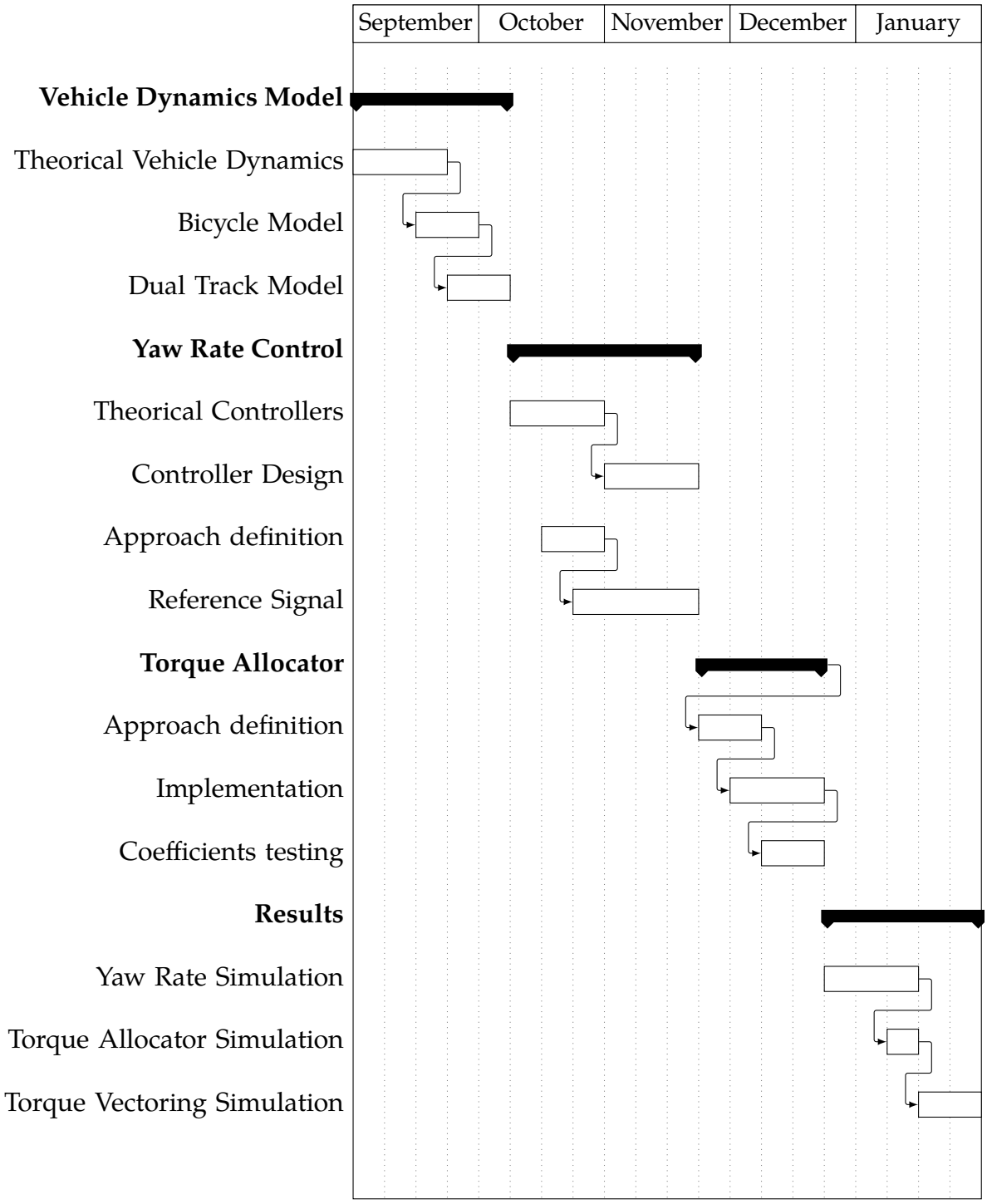


Table 1: Workflow Gantt Chart.

2 Formula Student

2.1 Introduction

Formula Student is an international automotive engineering competition where students from various universities design, build, and compete with their own single-seater racing vehicles. Founded in 1981 in the United Kingdom, the competition has grown to include events worldwide, attracting thousands of participants each year. The main objective of Formula Student is to stimulate the development of both practical and theoretical engineering skills, as well as promote innovation and collaboration between students and the automotive industry.

The student competition has transcended borders, giving rise to competitions all over the world. Some of the most important ones include Formula SAE in the United States, Formula Student Germany and Formula Student Japan. These events draw teams from renowned universities and technical colleges, providing participants with a global experience and the opportunity to compete at an international level.

Over the years, Formula Student has experienced a significant evolution, reflecting advancements in the automotive industry as well as changes within the competition itself. As technology has progressed, the vehicles designed for the competition have transitioned from an internal combustion engines to incorporating an increasing presence of electric propulsion systems in 2010, leading to the creation of the Electric Class. This adaptation not only mirrors the industry's shift towards electric mobility but also challenges students to be at the forefront of automotive innovation.



Figure 2.1: CAT15x prototype in FS Spain. August 2023.

Furthermore, Formula Student has expanded its scope with the introduction of new categories, such as the Formula Student Driverless in 2017. This challenges teams to design and build autonomous vehicles, highlighting the significance of autonomous driving technology in both the current and future automotive industry. These new cat-

egories reflect the ongoing evolution and relevance of Formula Student in the landscape of automotive engineering.

The latest change in this competition was made in January 2022 by Formula Student Germany, creating the Driverless Cup category. This category requires teams to have one vehicle for both the manual and driverless categories, whether it be combustion or electric.

2.2 Competition Event

In order to participate in the different competitions, several phases must be completed. First of all, each competition has its own quiz, which takes place in January, to select which teams will participate in them. After that, once in the competitions, technical inspections, both static and dynamic, must be passed.

Once all the inspections to ensure that your vehicle complies with the regulations have been passed, the competition begins. First, static events are held, followed by dynamic events which will be explained below. The following figure shows some of the different competitions, and their locations.



Figure 2.2: Formula Student Events Map.

Technical Inspections

As mentioned before, the competition has its own rules [5] and handbook [4], which mainly determine what a Formula Student car needs to be safe enough to be driven. To determine if any car is rules compliant, every team has to pass some tests carried out by the competition technical inspectors:

- **Pre-Inspection**

  It is a simple revision to bear out that all the driver equipment presented by a team is rules compliant and in a good state together with the tires and rims and the fire extinguishers.

- **Electrical Inspection**

  The main objective of this part is to determine if a car is electrically safe and rules compliant with all the electrical points such as the shut-down security circuit, all the electrical cables or the security firewalls.

- **Accumulator Inspection**

  The accumulator, its charger and basic tools for the treatment of both, will be analysed by the scrutineers.

- **Mechanical Inspection**

  This inspection is designed to determine if the car is mechanically safe and follows the regulations. Many items are reviewed: the monocoque, suspensions or the impact attenuator among others.

- **Diverless Inspection**

  Evaluates autonomous vehicle capabilities, including perception systems and decision-making algorithms. It ensures compliance with safety regulations for safe, human-independent operation.

- **Tilt Test**

  The tilt test is be conducted to assess a vehicle's stability, particularly during cornering or other dynamic maneuvers. To do so, the car is laid on a platform which raises from one side taking the car until 60º with the floor horizontal plane.

- **Rain Test**

  The objective of this test is to determine if all the electrical components are well protected from water and, a part from that, to determine if the high voltage and low voltage circuits are correctly isolated. To do so, the car is sprayed with water during 120s and left 120s more without being sprayed to see if the isolating sensor calculates a resistance below 250 Ω/V. If it does not do so, the test is passed.

- **Brake/EBS Test**

  The brake test is to determine if a car is able to block the wheels while braking and if it is stable during that maneuver. To validate a suitable braking, the car accelerates in a straight line, brakes at a determined distance and the competition staff determine if the maneuver fulfilled the requirements to pass. The EBS test is the same but in Driverless Class.

Score

As in all competitions, the Formula Student competition has its own scoring method. Both static and dynamic events are taken into account. The maximum score achievable is 1000 points in the Electric Class if you compete in the Driverless Cup, 850 points if you only compete manually, and 600 points in the Driverless Class.

	CV & EV	DC
Static Events:		
Business Plan Presentation	75 points	-
Cost and Manufacturing	100 points	-
Engineering Design	150 points	150 points
Dynamic Events:		
Skidpad	50 points	-
Driverless (DV) Skidpad	75 points	75 points
Acceleration	50 points	-
Driverless (DV) Acceleration	75 points	75 points
Autocross	100 points	-
Driverless (DV) Autocross	-	100 points
Endurance	250 points	-
Efficiency	75 points	-
Trackdrive	-	200 points
Overall	1000 points	600 points

Figure 2.3: Score table for the different FS Classes.

Static Events

- **Business Plan Presentation**

  The team has to present a business plan in which the Formula Student prototype has to be included, showing that it can be profitable. A team of judges are going to value it.

- **Cost and Manufacturing**

  All the materials and processes needed to build the car have to be exposed together with its corresponding costs. A team of technical judges are going to value it and ask for any doubts or how possible modifications would affect the final cost.

- **Design**

  The decisions taken until the final state of all the decisions taken on the car have to be presented. The exposing members have to explain to a group of technical judges all the considered options during the design process and what they know about the prototype.

Dynamic Events

- **Acceleration**

  The car has to run over a 75m long straight as fast as possible. Traction and power controls are crucial, as the aim is to do it as fast as possible without exceeding the limit power of 80kW determined by the rules.

- **Skid-Pad**

  The car has to turn twice right and twice left (in this order) around the track presented in the following figure. This event is taken to challenge the stability while cornering and also while making a fast direction change of the Formula Students cars.

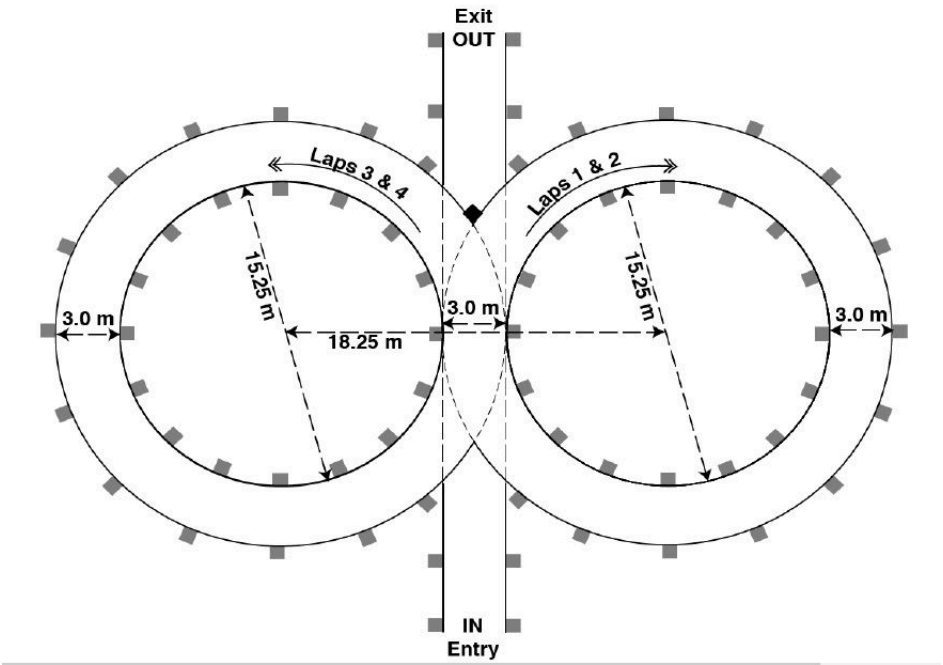


Figure 2.4: Skid-Pad track scheme.

- **AutoX**

  The aim of the AutoX or Autocross is to make the fastest possible lap around a circuit determined by each competition event. The total length of the track tends to be around 1km and it has to fulfill some limitations established in the regulations such as minimum turning radius or track width.

- **Endurance**

  The main objective is to finish a 22km run around the same or similar circuit of the AutoX event. This is the toughest test as it evaluates the reliability of all the components of the car and the capacity of maintaining the pace. There are several cars on the track at the same time, although wheel-to-wheel running is not allowed. For that reason there are zones where the car that is going to be passed has to go through to allow the car behind it to pass in case it is faster.

- **Efficiency**

  This is a test inside the Endurance itself which evaluates the efficiency of the cars in terms of energy usage.

  At the beginning and a the end of every event there is a pre and post scrutineering. At the pre-inspection, the judges determine if a car is still rules compliant and confirm that non-allowed changes have been done since the inspections. On the other hand, at the post-scrutineering the judges analyze that all the systems are in good state and if there have been any reliability problem, which would suppose a disqualification of a run or of that test itself.

- **TrackDrive**

  It consists on an endurance event of ten laps to the same track as the autocross event in the Driverless Class. This fact makes possible to save the autocross map, and compute the optimal trajectory offline. Then, at the time of the event, in-map localization is needed to localize the car in the track and the loaded map.

2.3 BCN eMotorsport

BCN eMotorsport is a Formula Student team formed by university engineers from different faculties of the Universitat Politècnica de Catalunya (UPC Barcelona). It was founded in 2007 under the name ETSEIB Motorsport with a combustion prototype, the CAT01, being the first Formula Student in Spain.

In the following years, the goal was to optimize the car's design. The year 2011 marked an important turning point in the team's history, as the decision was made to leave behind combustion engines and enter the world of electric vehicles with the CAT05e prototype. The team has evolved over the years, and in 2018, the first autonomous Formula Student car in Spain was created, developed by Driverless UPC. That same year, the ETSEIB Motorsport team incorporated a motor in each wheel.

Subsequently, in 2020, both Driverless UPC and ETSEIB Motorsport joined forces to form a single team, BCN eMotorsport, with the prototypes CAT13e and CAT13d. In 2021, the team decided to join the Driverless Cup, so the CAT14x was the team's first hybrid car. The last season, the CAT15x, represented the evolution of the CAT14x and competed in Formula Student Italy, Formula Student Spain, and Formula Student Germany.



Figure 2.5: CAT15x's Formula Student Team.

3 Vehicle Dynamics

3.1 Introduction

Vehicle dynamics [15] is a fundamental field in automotive engineering that focuses on the study of the motion and behavior of moving vehicles. Understanding vehicle dynamics is necessary to understand and develop any control system, especially those aimed at enhancing vehicle performance.

Different kinematic and dynamic variables, such as velocities, forces, and displacements, are key points for analyzing vehicle dynamics. To do this, it is essential to define proper reference axes. In the automotive sector, it is common to define reference axes as shown in the following figure. These axes have three possible movements in space: Yaw, Pitch, and Roll.

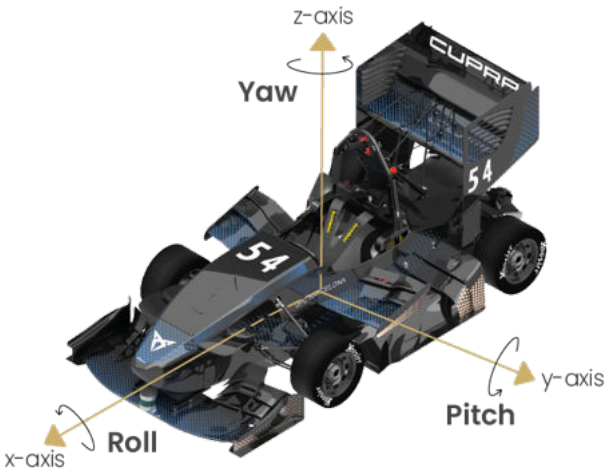


Figure 3.1: Vehicle Reference Axis.

In this thesis, it is important to use a 4WD vehicle as we aim to push the car to its physical limits in order to better understand its state. We will employ a two-track nonlinear model, which will be examined in more detail in Section 4.2.

3.2 Load Transfer

A very important feature in car design is the center of gravity. Each vehicle has a weight distribution determined by its internal components, which means that even in a static state, there may already be an asymmetric distribution of vertical force on each axis and even between right and left, although this latter case is much less common.

Load transfer [11] is an important aspect to consider, particularly in vehicles that test the limits of vehicle, such as competition vehicles, for example, Formula/Rally-type cars. This is because, along with the vehicle's movements, the weight distribution changes, and each tire receives a different vertical load that directly affects longitudinal and lateral forces it can handle.

Load transfer occurs due to dynamic forces during acceleration, deceleration, cornering, and as a result of speed. Under these conditions, the load on the wheels can shift, impacting the behaviour and stability of the vehicle.

3.2.1 Longitudinal Transfer

Longitudinal load transfer is caused by acceleration or deceleration forces. The effect of this depends on both the longitudinal forces and the position of the previously mentioned center of gravity, both in terms of x position and height.

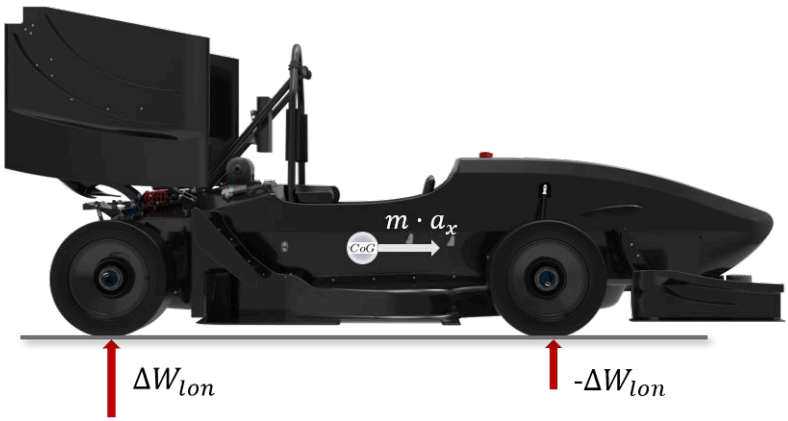


Figure 3.2: Longitudinal load transfer.

As can be observed in Figure 3.2, it is possible to establish the relationship using a free-body diagram, the following expression for longitudinal load transfer can be deduced:

$$\Delta W_{lon} = \frac{h_{CoG} m a_x}{l} \tag{1}$$

Where a_x $[m/s^2]$ is the longitudinal acceleration, m $[kg]$ is the mass, h_{CoG} $[m]$ is the height of the center of gravity, and l $[m]$ is the wheelbase of the vehicle.

3.2.2 Lateral Transfer

Lateral load transfer, on the other hand, is induced by forces acting during cornering maneuvers. Its impact is influenced by the lateral forces and the position of the center of gravity height, similar as mentioned earlier.

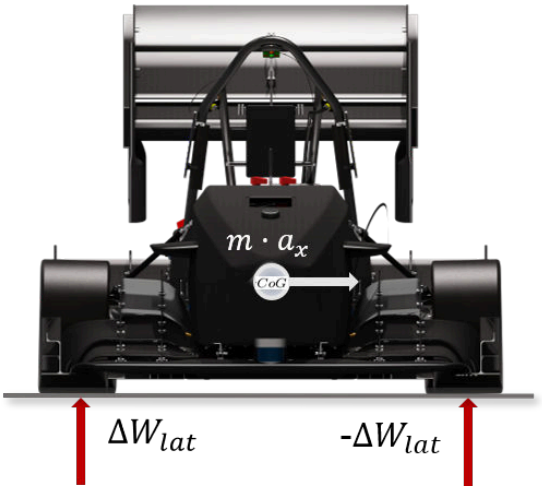


Figure 3.3: Lateral load transfer.

Using Figure 3.3, it is possible to establish the relationship using a free-body diagram, the following expression for longitudinal load transfer can be deduced:

$$\Delta W_{lat} = \frac{h_{CoG} m a_y}{t_i} \tag{2}$$

Where a_y $[m/s^2]$ is the lateral acceleration, m $[kg]$ is the mass, h_{CoG} $[m]$ is the height of the center of gravity, and t_i $[m]$ is the track of the axis, being the rear axis $(i = r)$ and the front one $(i = f)$.

However, since the motion is related to roll motion, even though the roll angle is neglected, the height of the roll center and the stiffness distribution between axles are taken into account [12]. Next, the Figure 3.4 will shown the diagram of an axle and the related equation for that axle, in this case, the front axle:

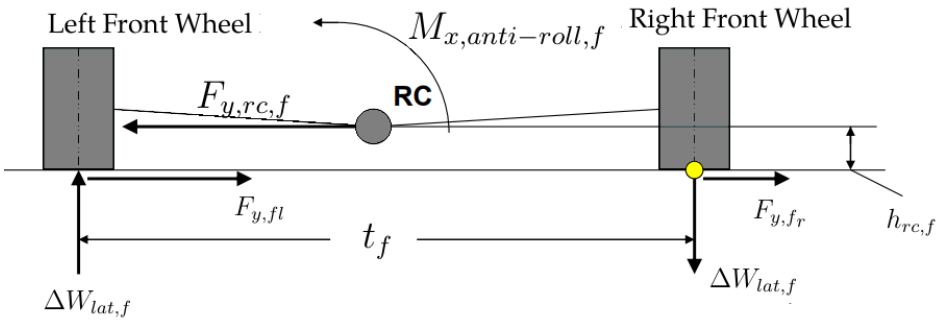


Figure 3.4: Lateral transfer load using roll motion.

$$\Delta W_{lat,i} = a_y \cdot \left(\frac{m_i h_{rc,i}}{t_i} + \frac{K_{roll,i}}{K_{roll,f} + K_{roll,r}} \cdot \frac{m_f(h_{CoG} - h_{rc,f}) + m_r(h_{CoG} - h_{rc,r})}{t_i} \right) \quad (3)$$

where a_y $[m/s^2]$ is the lateral acceleration, m_i $[kg]$ is the mass of the i axle, $h_{rc,i}$ $[m]$ is the height of the roll center of the axle i, $K_{roll,i}$ $[N/m]$ and t_i $[m]$ is the track of the i axle, being the rear axle$(i = r)$ or the front one $(i = f)$.

3.2.3 Aerodynamics Load

Aerodynamic force is the impact that air exerts on a moving vehicle. This force can affect the speed, stability, and efficiency of the vehicle, and it results from the interaction between the vehicle and the airflow around it. The goal is to design vehicles with efficient aerodynamic profiles to minimize resistance and enhance performance.



Figure 3.5: Aerodynamics forces.

This interaction can be differentiated into two distinct forces. The lift force is the aerodynamic force that acts vertically, generating lift or downforce. In vehicles, it is used to generate downforce and provide a higher vertical force than without the aerodynamics components and thanks to this the tire can support more force. On the other hand, the drag force is the resistance that opposes the object's motion in the direction of the airflow.

$$F_{lift} = \frac{1}{2}\rho A C_l v_x^2 \quad \text{and} \quad F_{drag} = \frac{1}{2}\rho A C_d v_x^2 \quad (4)$$

where ρ $[kg/m^3]$ is the density of the air, A $[m^2]$ is the cross section, v_x $[m/s]$ is the longitudinal velocity, C_l $[-]$ is the coefficient of lift and C_d $[-]$ is the coefficient of drag.

3.3 Tire Dynamics

Tires are a crucial component in studying and understanding vehicle behavior, being a fundamental element in both the driving experience and the accuracy of simulation models. Tires are the only point of contact between the vehicle and the road, playing an essential role in the safety, performance and efficiency of the vehicle.

The Pacejka Magic Formula [7] is a popular mathematical model that accurately describes how tires behave based on factors like slip angle and load. It's a key tool in vehicle dynamics, helping design vehicles, develop control systems, and run simulations. Its versatility in different conditions makes it crucial for predicting tire forces and improving the realism of virtual vehicle models.

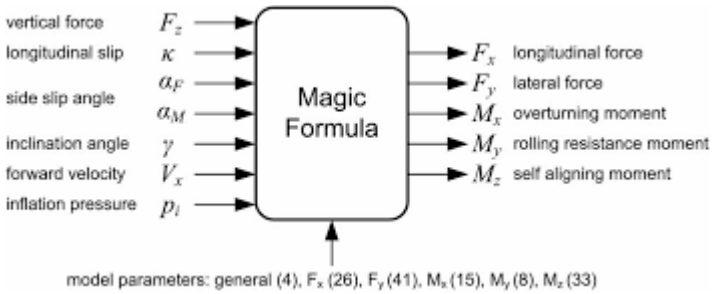


Figure 3.6: Inputs and outputs from Magic Formula.

The general Magic Formula depends on lots of variables and parameters as can be seen in 3.6. From now, a simplified version will be considered taking into account less inputs and outputs as can be seen in Figure 3.7.

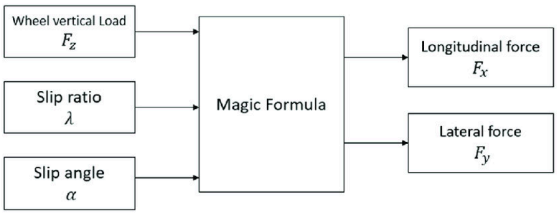


Figure 3.7: Inputs and outputs from a simplified Magic Formula.

The different variables corresponds for each tires, where F_z $[N]$ is the vertical load, λ $[-]$ is the slip ratio, α $[rad]$ is the slip angle, F_x $[N]$ is the longitudinal force and F_y $[N]$ is the lateral force, both supported by the tire.

3.3.1 Pure Longitudinal Force

The longitudinal force of a tire, which drives or slows a vehicle, is generated at the tire-road contact and the pure longitudinal force takes the slip angle as zero, this value are explined in Section 3.3.2. It comes from the tire's grip on the road surface. During acceleration the driven wheel rotates faster than a free-rolling wheel while during braking

the driven wheel rotates slower, so the difference of rotational speed is defined as slip ratio.

$$\sigma_i = \frac{w_i R_w - v_{x,i}}{max\left(w_i R_w, v_{x,i}\right)} \quad \text{where} \quad i = fl, fr, rl, rr \tag{5}$$

where w_i $[rad/s]$ is the angular veloicty of the wheel, R_w is the wheel radius, $[m]$ is the vehicle mass and $v_{x,i}$ $[m/s]$ is the longitudinal velocity of the wheel.

Slip ratio is defined as the equation 5, where wR_w will be higher during acceleration and v_x will be higer during braking. In the following scheme 3.8 will see the different variables that interacts in longitudinal force.

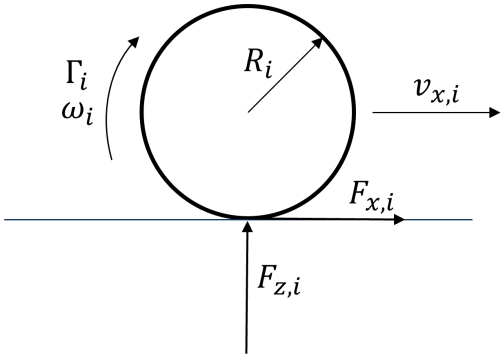


Figure 3.8: Slip Ratio Scheme.

In the scheme above, some ideal states are considered like a constant wheel radius and the tire rolling resistance are neglected. Considering the sum of momentum at the center of the wheel it can be deducted the following equatoin:

$$F_{x,i} = \frac{1}{R_w}\left(\Gamma_i - J_{w,i}\dot{w}_w\right) \quad \text{where} \quad i = fl, fr, rl, rr \tag{6}$$

where Γ_i $[Nm]$ is the wheel torque, R_w $[m]$ is the wheel radius, $J_{w,i}$ $[kg\ m^2]$ is the inertia of the wheel and $\dot{w}_w$ $[rad/s^2]$ is the angular acceleration of the wheel.

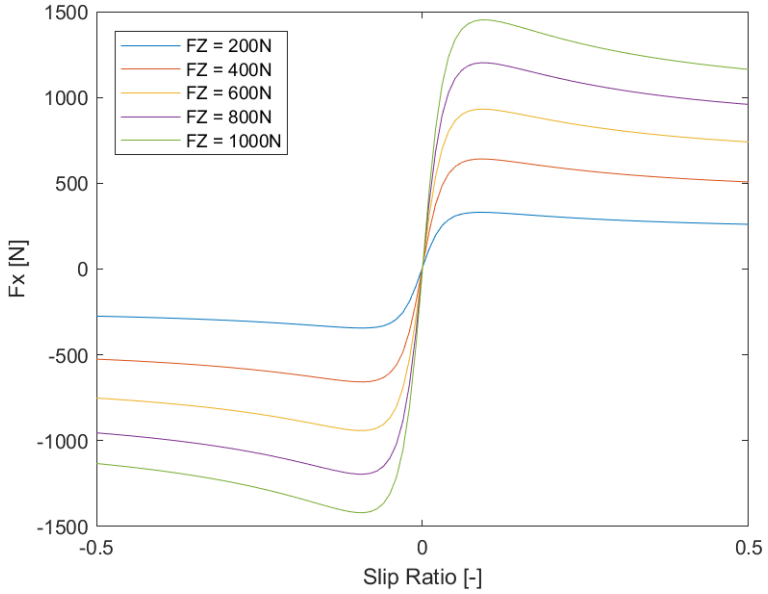


Figure 3.9: Fx vs Slip Ratio.

As shown in the figure above, some slip ratio is necessary to develop the desired longitudinal force and greater vertical load also allows more longitudinal force. Concerning the spline, in the initial segment, the relationship between the longitudinal force and the slip ratio is linear, known as the longitudinal stiffness of the tire. Following this segment, there is a peak where the maximum allowable longitudinal force is reached, after which the force stabilizes at a constant value.

3.3.2 Pure Lateral Force

The lateral force of a tire, responsible for steering and cornering, occurs at the tire-road interface, generated by the tire's grip on the road surface. When a vehicle steers, the lateral force comes into play, causing the tire to move sideways.

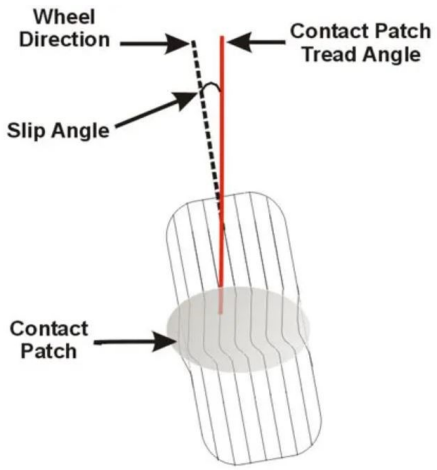


Figure 3.10: Slip Angle Scheme.

As can be seen in the figure above, the slip angle defines the difference between the actual direction of the wheel and the direction it would have without lateral force. During cornering, the slip angle is essential, as the wheel deviates from its intended direction as the following equation:

$$\alpha_i = \delta_i - arctan\left(\frac{v_{y,i}}{v_{x,i}}\right) \quad \text{where} \quad i = fl, fr, rl, rr \tag{7}$$

where, δ_i $[rad]$ is the wheel steering $v_{x,i}$ $[m/s]$ is the wheel longitudinal velocity and $v_{y,i}$ $[m/s]$ is the wheel lateral velocity.

In the following Figure 3.11, some slip angle is necessary to develop the desired lateral force and greater vertical load also allows more lateral force. Concerning the spline, in the initial segment, the relationship between the lateral force and the slip angle is linear, known as the cornering stiffness of the tire. Following this segment, there is a peak where the maximum allowable lateral force is reached, after which the force starts decreasing and stabilizing at a constant value as happens with longitudinal force.

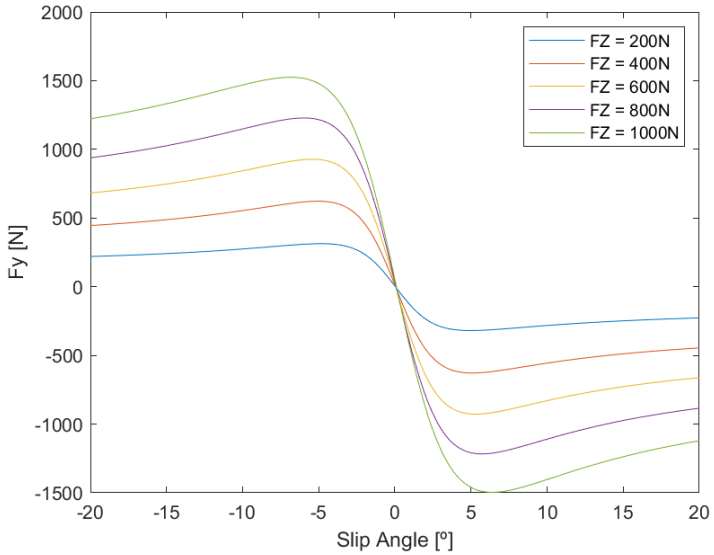


Figure 3.11: Fy vs Slip Angle.

3.3.3 Friction Circle

In previous sections, forces were studied as isolated cases of pure force. However, when considering the longitudinal force in relation to the slip angle or when considering the lateral force in relation to the slip ratio, an entirely different scenario emerges.

On the one hand, in the longitudinal case, fixing the slip ratio and varying the slip angle can be observed the following figure and it is important to highlight it is a nonlinear relationship.

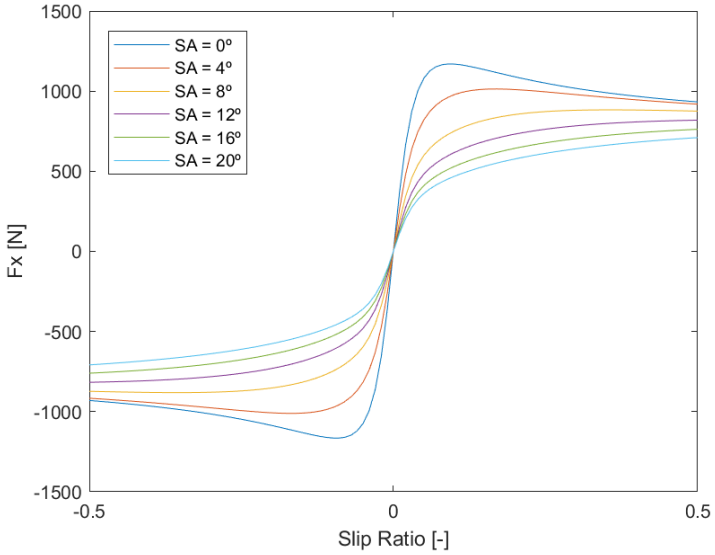


Figure 3.12: Fx vs Slip Ratio vs Slip Angle.

On the other hand, in the lateral case, fixing the slip angle and varying the slip ratio can be observed the following figure and as the longitudinal case is a nonlinear realtionship.

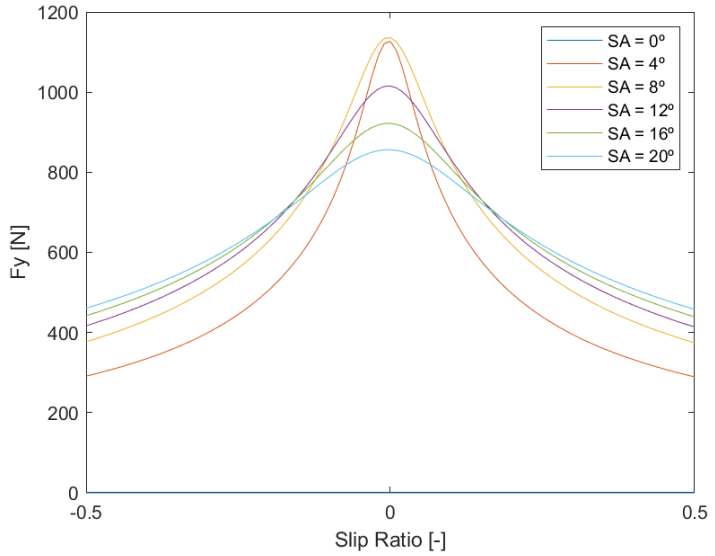


Figure 3.13: Fy vs Slip Ratio vs Slip Angle.

Finally, combining both cases, the friction cricle of a tire is defined. This circle allows to known the exact conditions of the tire and its limits only knowing the normal load, the slip ratio and the slip angle. There are several variables that define those limitations as the pressure, temperature, camber or velocity among others but it will make more complex to define the friction circle.

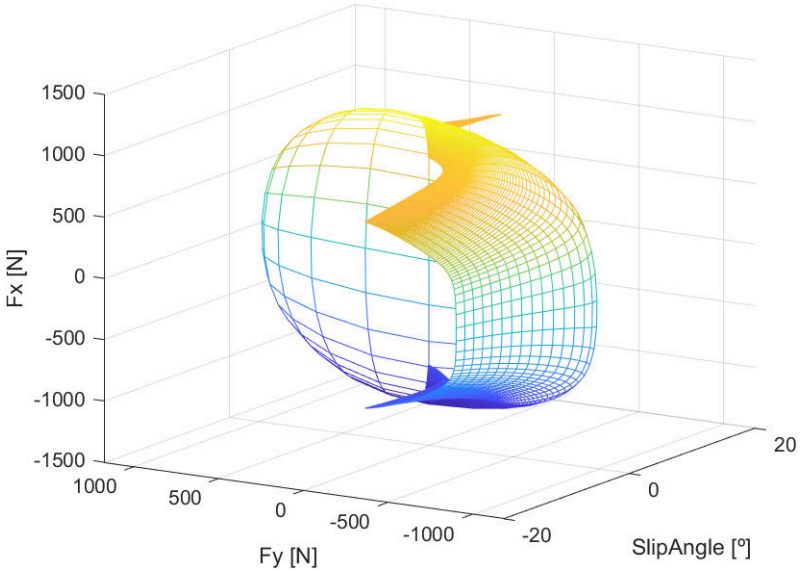


Figure 3.14: Friction Circle.

3.3.4 Ackermann Steering Geometry

The first steering geometry used was parallel steering, where the inner and outer wheels have the same angle. This was achieved by an easy method in the steering system, as the steering tie rod is perpendicular to the steering arm. This method causes the inner wheel to follow a smaller turning radius than the outer wheel, leading to excessive wear and reducing the lifespan of the tires. The following figure illustrates an example of what parallel steering looks like.

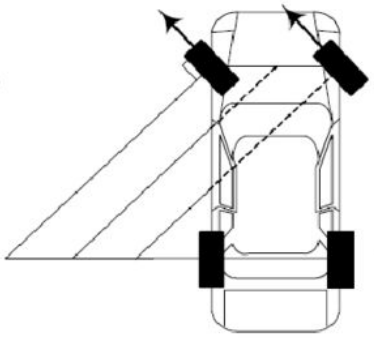


Figure 3.15: Parallel Steering System.

Ackermann steering geometry is a design principle applied to the steering systems of vehicles to enable smoother and more precise turning. Its principle involves turning the inner wheel at a different angle while maintaining a smooth, continuous arc during a turn. Two types of Ackermann steering are shown in Figure 3.16.

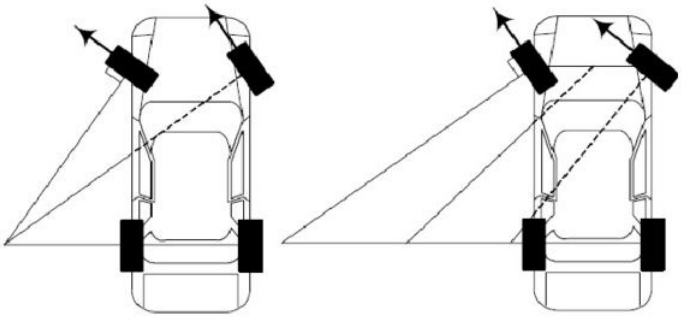


Figure 3.16: Ackerman Steering System.

In one hand, the car on the left uses Pro-Ackermann, where the inner wheel has a greater angle than the outer wheel. This setup is designed to enhance stability and improve handling during cornering. In the other hand, the car on the right uses Anti-Ackermann steering, where the outer wheel has a greater angle than the inner wheel. This configuration is often used in certain vehicles or situations to improve maneuverability at lower speeds or in tighter spaces as in the case in the Formula Student.

3.4 Vehicle Behaviour

Understanding how a vehicle behaves on the road involves two important concepts: understeer and oversteer. These terms describe how a car responds when taking a turn. Understeer happens when you turn the steering wheel, but the car does not follow as sharply as you had expected. It occurs when the front tires are struggling for grip and the car feels like it is pushing forward through the turn. On the other side, oversteer occurs when the rear tires lose traction before the front ones. This causes the back of the car to slide outwards, almost as if it is trying to overtake the front and the car feels like it is losing control.

These behaviors are influenced by various factors, as how much weight is on each tire, the condition of the tires, how the suspension is set up and even the road conditions, all play a part. Knowing about understeer and oversteer is crucial for drivers because it helps them anticipate and handle these situations better. By understanding how these behaviors work, drivers can react appropriately, making driving safer and more controlled.

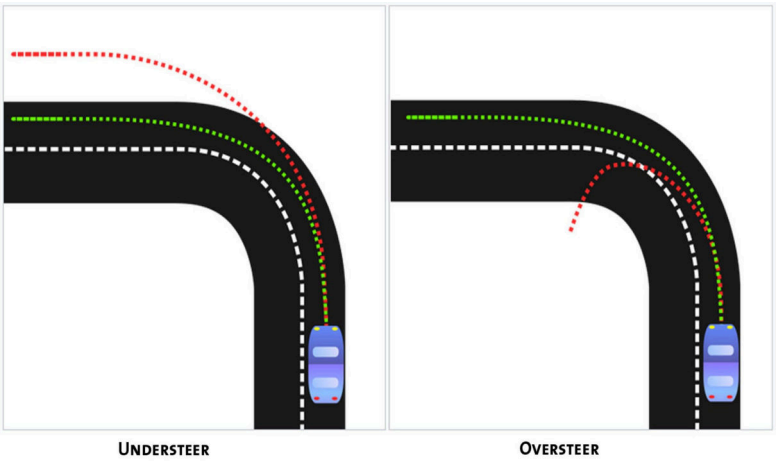


Figure 3.17: Understeer and Oversteer representation.

3.5 Vehcicle Data

3.5.1 Vehicle Comunication

Car communications systems has become vital interactions between vehicles, enhancing safety, efficiency and convenience on the roads. Car communication is about creating a network that enables real-time data exchange, enabling vehicles to make informed decisions. While commercial cars already have their own communication systems, Formula Student cars have also adopted this technology.

CAT15x uses two different methods. The first one and the most common on actual vehicles is Controller Area Network (CAN), serving as a communication protocol that enables interaction among various parts of a car. In CAT15x is used to read or send data from the different sensors or ECUs.

The second one is Robotic Operating System (ROS), working as a communication protocol that facilitates interaction between different components of a robotic system. Developed to simplify the communication of robotic software, ROS enables efficient communication among various nodes, allowing robots to process information, make decisions and perform tasks. In CAT15x, all different systems and planning algorithms, both manual and autonomous, uses ROS to communicate as individual nodes.

To bridge these two communication protocols, a Processing Unit (PU) serves as the main ECU of the car. This centralization in the PU allows the entire car's systems to be coordinated. However, this setup also have its cons due to the significant computational load it entails. The following figure illustrates the communication scheme of the car, demonstrating how the different algorithms and devices communicate with each other.

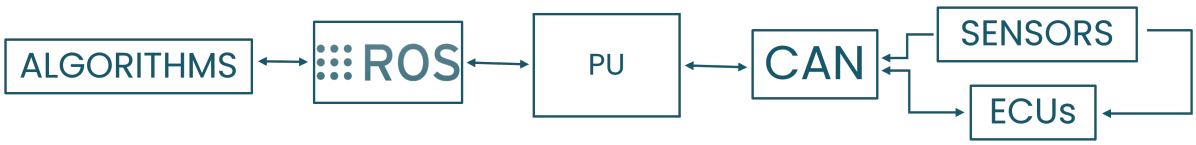


Figure 3.18: Car communication scheme.

3.5.2 Sensors and actuators

Lineal Position Sensor

The linear position sensor is used for three different functions.

- The first one is to mesure the position of the pedals, one sensor for the brake pedal and two for the throttle.
- The second functionality for this sensors is to mesure the steering rack position and with it know the steering wheel angle.
- The last one is used to know the compression of the suspension springs. From this, the vertical loads for each wheel can be determined. In this case

In both first and second case the MLP-25 Linear Displacement Sensor were used and in the third case were used the CLS0950 Ultra-slim.



Figure 3.19: MLP-25 Linear Displacement Sensor.

Inertial Navigation System - INS

The INS used is the VN-300 from VectorNav. It is made up of an IMU and a GPS so it can be used for different objectives. The sensor is placed inside the PU box, under the driver seat, close tot he CoG of the CAT15x.

As mentioned before, it gives some of the most important mesurament needed for contol algorithms and it can be categorized into three groups:

- **Position:** The GPS gives the geographical position of the car, allowing us to know where the car is.

- **Acceleration:** The IMU have a 3-axis accelerometers giving a_x, a_y and a_z and a 3-axis gyros giving the orientation and angular velocity, being ϕ, ψ and $\dot{\psi}$ the ones used.

- **Velocity:** The main objective of the INS is to combine the IMU and GPS information to gives a more robust and accurate navigation, thus being able to give velocity for each axis v_x and v_y.

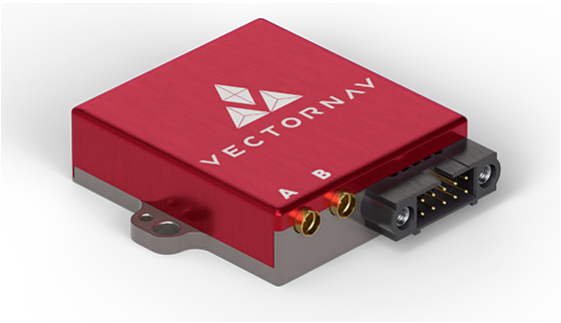


Figure 3.20: VectorNav VN-300 INS sensor.

Resolver

The resolver used is the RE-15 from LTN is a rotary electrical sensor designed to precisely measure rotational position and speed. It is necessary for the correct operation of the inverter and some other algorithms as traction and power control.



Figure 3.21: LTN RE-15 resolver.

Electrical Machines

The electrical machines used are TI085-052-070-04B7S from Fischer. Most of this motors are equiped in Formula Student cars. It is about engines capable of generating 29 Nm of torque and rotating up to 20,000 revolutions per minute. Due to the high revolutions per minute, the motors are connected to the wheel through a planetary transmission with a 9.6 ratio.



Figure 3.22: Fischer TI085-052-070-04B7S motors.

Inverter

The inverter used is the MOBILE DCU 60/60 from Lenze. The Lenze DCU 60/60 inverter uses the information provided by the RE-15 resolver to precisely control the speed and position of two of the Fischer motor, allowing for highly precise and efficient control. Since you have 4 motors, you need two inverters to control all the engines.

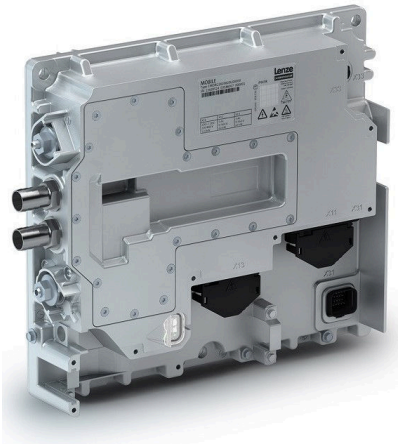


Figure 3.23: Lenze MOBILE DCU 60/60.

Energy Meter

To know information about the voltage, intensity, power and energy consumption for the CAT15x, a Isabellenhüte IVT-MDOULAR energy meter sensor is used.



Figure 3.24: Isabellenhüte IVT-MDOULAR energy meter sensor.

3.5.3 Data Adquisition

In motorsport, data acquisition is a crucial aspect in competition because its use brings several advantages. One of the main benefits of colleting data while the car is racing is the ability to verify the vehicle's behavior. Another significant advantage is using this data both to design controllers, such as torque vectoring or traction control, and to check afterward whether the behavior using these controllers is correct, aiding the vehicle's performance.

To manage all the information, there's a need for a way to store the sensor or message value and its respective time. Since CAT15x processes all messages in ROS, it offers a function called rosbag to save all desired values and later replay them exactly as in the vehicle. This is highly useful for reproducing errors and fixing them without the need for making a test again. Another method used in data collection is storing the values in a CSV file, where information is written as messages are sent through ROS.

Obtaining and understanding sensor data in motorsport is essential [10] to comprehend the vehicle's behavior and detect issues. The Data Analysis Pro software (GEMS Performance Electronics) streamlines the import and organization of data, enabling customized visualization based on the team's objectives.



Figure 3.25: DataAnalysisPro Example.

4 Vehicle Dynamics Model

The vehicles models are both mathematical and physics representations that calculate the behaviour of the car in different conditions. These models are fundamental for the design and development of controllers, as they allow simulating and know to understand the behaviour before various inputs and scenarios. From simple models to a more complex one as multi-body systems [9], these models are essentials tools for control systems engineering to design strategies that improves the stability, efficiency and performance of the vehicle.

4.1 Bicycle Model

The most common vehicle model used and a basic one is the bicycle model. This name comes from the fact that it simulates the behavior of the vehicle in a similar way as a bicycle. This model consists in two axis, the front and rear axis, each one represented by only one wheel and join by a structure that represent the chassis of the vehicle.

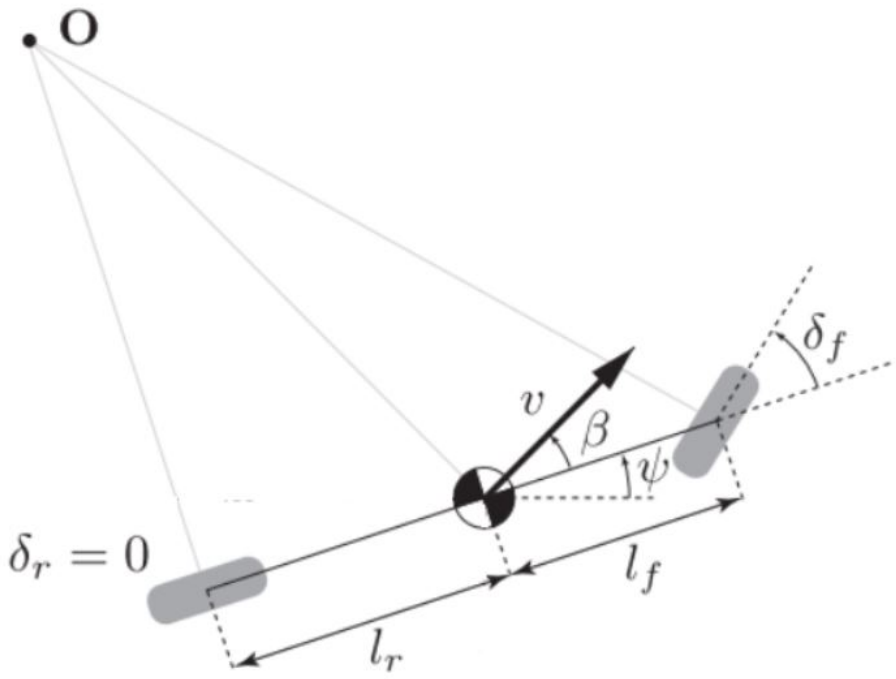


Figure 4.1: Bicycle Model Scheme Representation.

Even though is a simplification, this model is used to design different controllers, as it allows to understand some important aspects of the vehicle dynamics, like the direction, the stability and the response to some inputs. The model us defined as:

$$\begin{aligned} \ddot{\psi} J_v &= l_f F_{y,f} - l_r F_{y,r} + M_{z,TV}, \\ \dot{\beta} &= \frac{F_{y,f} + F_{y,r}}{mv} - \dot{\psi} \end{aligned} \tag{8}$$

where $\ddot{\psi}$ $[rad/s^2]$ is the yaw acceleration, J_v $[kg\ m^2]$ is the vehicle inertia, $M_{z,TV}$ is the additional yaw moment, $\dot{\beta}$ $[rad/s]$ is the vehicle slip angle rate, m $[kg]$ is the vehicle mass and $\dot{\psi}$ $[rad/s]$ is the yaw rate.

As the equation takes into account the slip angle of the vehicle's center of gravity, a simplification will be used to calculate the slip angle of each wheel in order to determine the forces on the axes:

$$\begin{aligned} \alpha_F &= \beta + \frac{l_f \dot{\psi}}{v} - \delta, \\ \alpha_R &= \beta - \frac{l_r \dot{\psi}}{v} \end{aligned} \tag{9}$$

where α_i $[rad]$ is the slip angle of each wheel and v $[m/s]$ is the velocity of the vehicle.

4.1.1 LTV

A Linear Time Varying or LTV is a type of mathematical model which the relationships between variables or parameters change linearly over time. In contrast to a standard Linear Time Invariant or LTI model, where the relationship between the parameters remain constant, a LTV model allows the evolution of the variables. This is useful when the systems conditions constantly change. Another characteristic is the lineal complexity because although the changes of the parameters, the defined equations of the system are still lineal.

This model can be expressed using a state-space as:

$$\begin{aligned} x^+ &= A(\beta)x + B(\beta)u, \\ y &= C(\beta)x + D(\beta)u \end{aligned} \tag{10}$$

where

$$\begin{gathered} A = \begin{bmatrix} \frac{C_r(\beta)l_r - C_f(\beta)l_f}{J_v} & \frac{C_r(\beta)l_r - C_f(\beta)l_f}{mv^2} - 1 \\ \frac{-(C_f(\beta)+C_r(\beta))}{mv} & \frac{C_r(\beta)l_r^2 + C_f(\beta)l_f^2}{J_v v} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{C_f(\beta)}{mv} & 0 \\ \frac{C_f(\beta)L_f}{J_v} & \frac{1}{J_v} \end{bmatrix}, \\ C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \\ x = \begin{bmatrix} \beta \\ \dot{\psi} \end{bmatrix} \text{ and } \quad u = \begin{bmatrix} \delta_w \\ M_{z,TV} \end{bmatrix} \end{gathered} \tag{11}$$

where the inputs are δ_w $[rad]$ which is the wheel angle at the wheel and $M_{z,TV}$ $[Nm]$ is the additional yaw moment. The other variables are $C_i(\beta)$ $[N/rad]$ is the cornering stiffnes of the tire i and depends on the β $[rad]$,

In this case the parameter varying is the cornering stiffnes of each tire, which varies with its slip angle.

4.1.2 NLTV

Secondly, another version of the LTV is the Nonlinear Time Varying (NLTV) model, which is similar to the previous model but adds a higher level of complexity. In this case, the parameters do not vary linearly over time, making their behaviour unpredictable and unable to be expressed using a state-space.

This time, the parameter that varies is the x and y forces supported by the tyre, but this time it changes based on the vertical force and the slip angle of that axis. This increase in complexity makes the model more accurate in reality compared to the LTV. On the one hand, this initial model is accurate when the vehicle's behaviour is within the linear zone of the tyre, as discussed in Section 3.3. On the other hand, the NLTV model is also accurate when the tyre is in the non-linear zone, allowing us to more precisely determine the vehicle's limits, as depicted in the following graph.

This nonlinearity comes with a drawback, requiring the tyre model to be more accurate with reality. Any significant difference between reality and the model will render this nonlinear part inaccurate, making this complexity not worthwhile.

To employ this model, concepts and equations elucidated in the Bicycle Model, Long Transfer, and Pacejka will be used.

4.2 Dual Track NTV

After considering the previous simple model, as this project focuses on a 4WD vehicle with a motor on each wheel, a four-wheel traction car is being considered. It is true that the bicycle model already considers the vehicle's lateral dynamics, as well as the possibility of adding an additional yaw moment. However, by separately considering all four wheels, we get a bit closer to reality, as more aspects can be taken into account.

In this case, a Dual-Track model will be used, which involves the general dynamics of the vehicle from the Bicycle Model 4, but taking into account all four wheels, lateral load transfer 3.2.2, the x and y forces of each wheel 3.3, as well as the steering geometry 3.3.4 for the resolution of forces. Below is a schematic representation of the model.

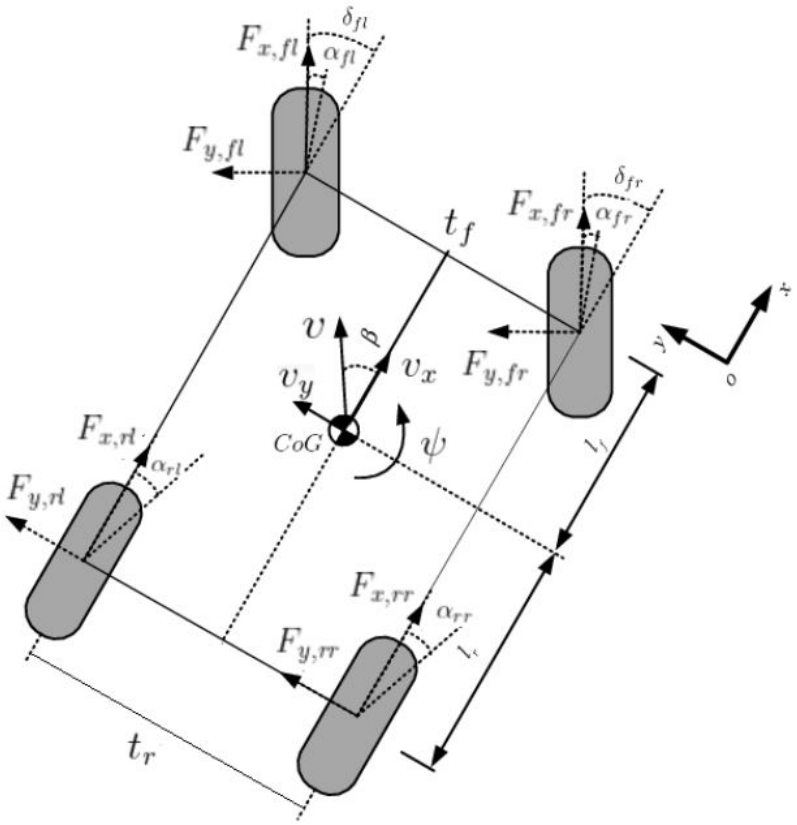


Figure 4.2: DualTrack Model Scheme Representation.

As mentioned earlier, four different concepts are used in this model. Below is a block diagram showing its inputs and outputs in the model.

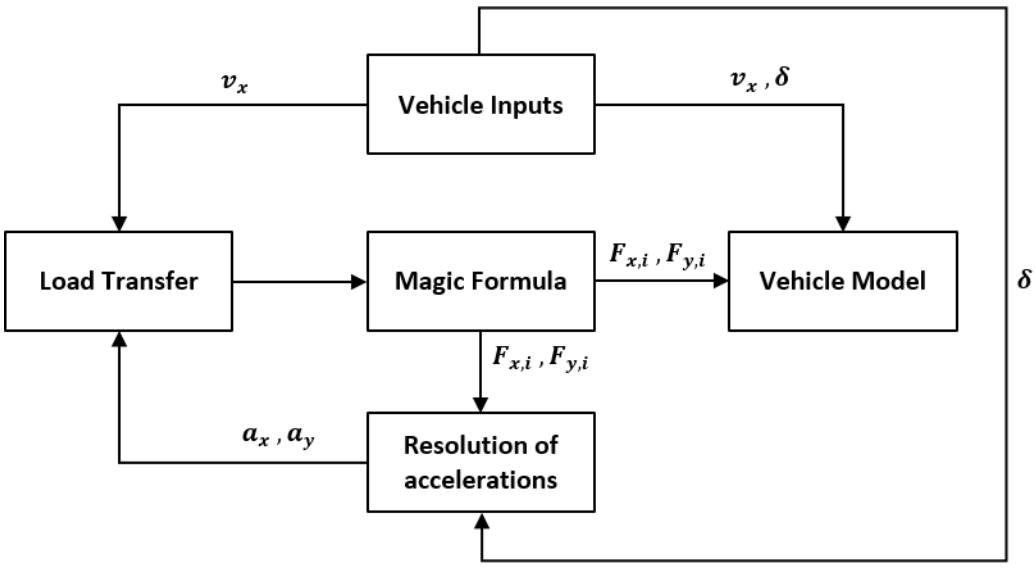


Figure 4.3: Diagram Dual Track Model.

In the load transfer the longitudinal and lateral dynamics are taken into account as well as the influence of the aerodynamic force. The accelerations decomposition only uses the front wheel steerings to get the forces in the axis reference of the vehicle.

4.2.1 Model Validation

The behavior of the vehicle model, to be useful, has to truly fit reality, especially if it will be used as a basis for designing a controller, as is the case. Therefore, in this section, its behavior will be validated in various different situations against the real behavior of the CAT15x.

To achieve this, the relative experimental error between the yaw rates of both vehicles will be calculated as follows:

$$\varepsilon_{rel} = 100 \frac{\sum_{i=1}^{n} |\dot{\psi}_i^R - \dot{\psi}_i^M|}{\sum_{i=1}^{n} |\dot{\psi}_i^R|} \tag{12}$$

where n is the total number of poits of the data vector, $\dot{\psi}_i^R\ [rad/s]$ and $\dot{\psi}_i^R\ [rad/s]$ are the yaw rates of the position i in the data vector, the one with the ^{R} is from the real data vector and the ^{M} is from the model data vector.

First of all, we will employ a very simple maneuver that provides insight into the lateral dynamics of the vehicle. The referred maneuver is the Ramp Steer. This is executed at a constant speed, and gradually, the wheel angle is increased with a constant gradient. In all graph legends, the percentage fit of the model versus the actual data will be displayed.

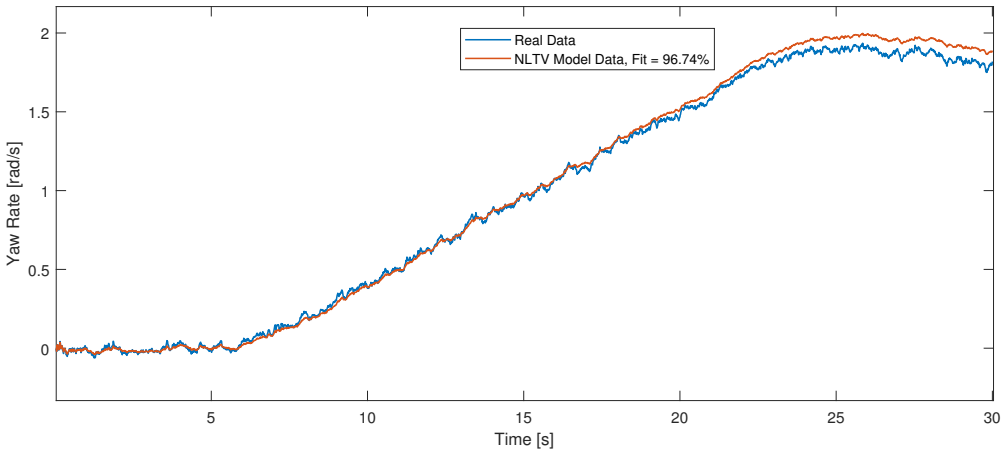


Figure 4.4: Comparation between NLTV Model and CAT15x Data on a Ramp Steer.

The second test conducted to verify the model and to add a bit more complexity to the maneuvers performed will be the SkidPad track, as defined in Figure 2.4.

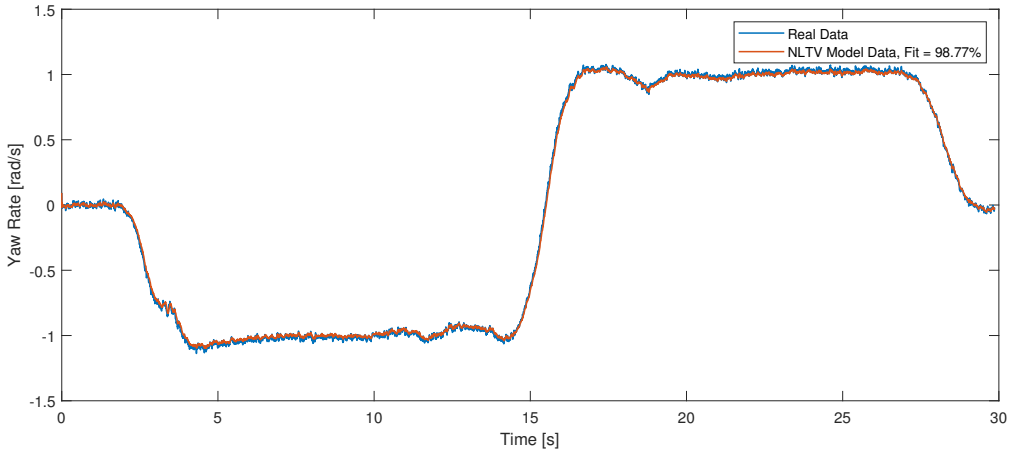


Figure 4.5: Comparation between NLTV Model and CAT15x Data on a Skid Pad.

Lastly, to conclude with the observation of the strengths and weaknesses of the model, the comparison of the yaw rate of the model against the CAT15x data recorded during the AutoX in the Formula Student Spain competition at the Circuit de Barcelona-Catalunya in August 2023. To highlight this competition, we secured the first position in the EV Class.

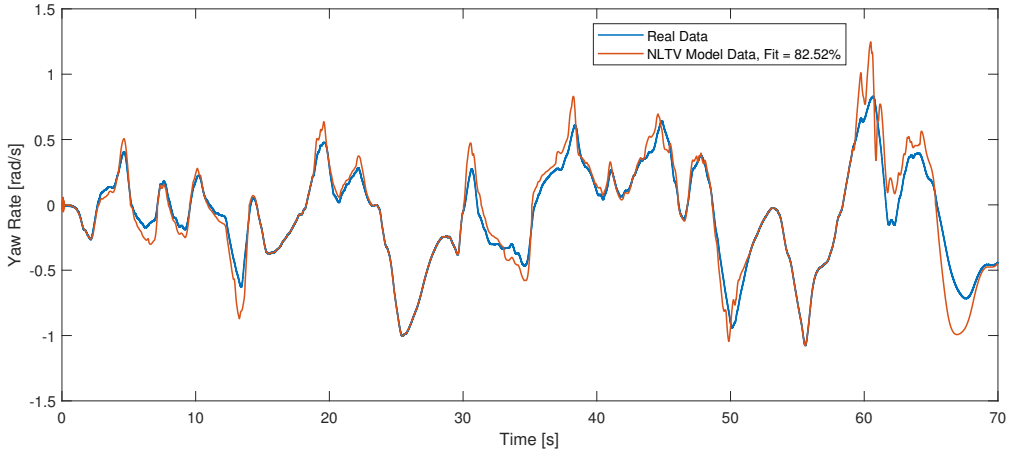


Figure 4.6: Comparation between NLTV Model and CAT15x Data on the FSS AutoX 2023.

To conclude the model validation, observing the results obtained in the three different tests, we can conclude that the model fits well with the behavior of the CAT15x.

It is true that in some cases, such as the final part of the Ramp Steer in Figure 4.4 or during some sections of higher yaw rate in the AutoX of Figure 4.6, there is more error between both vehicles. Nevertheless, the error is small enough to consider the results satisfactory.

4.3 VI-Grade Model

Finally, a simulation tool has been used in this project to test the designed control and tune it before using it on the CAT15x. This is crucial to progress in the controller development since testing to verify its performance is not always available, and using a commercial simulation tool brings the vehicle's behavior much closer to reality than using your own.

The simulation tool used is VI-Grade, a leading company in simulation for the automotive industry. They provide advanced technology and software to simulate and test vehicles in virtual environments, enabling manufacturers to optimize vehicle development and reduce costs.

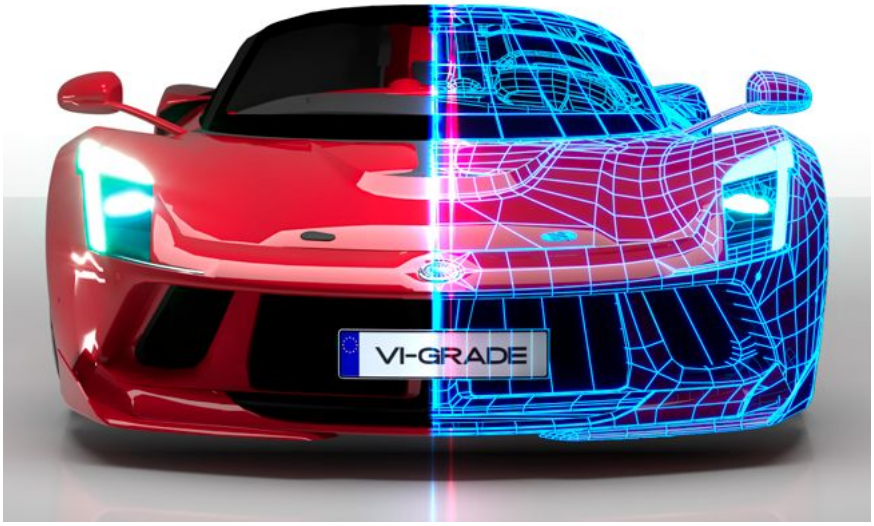


Figure 4.7: VI-Grade Simulation Tool.

Among VI-Grade's tools, VI-CarRealTime [14] has been used, which is a key tool for real-time simulation of vehicles. This software allows modeling and simulating the dynamic behavior of vehicles in a virtual environment, replicating real driving conditions with high precision. It's used by car manufacturers and research centers for real-time testing and analysis, aiding in enhancing driving dynamics, safety, and vehicle performance.

VI-Grade's vehicle model is based on a 14 DOF multibody approach, of which 6 of them represent linear displacements and rotations along the x, y, and z axes and the other 8 DOF are 2 for each wheel which consist of the vertical motion of the wheel and the wheel spin. This model describes vehicle dynamics considering aspects such as suspension, steering geometry, mass distribution, tire-road interaction, and other factors influencing vehicle behavior during driving. By simulating this model, engineers can analyze and optimize vehicle performance in various driving conditions and scenarios, crucial in designing safer and more efficient vehicles.

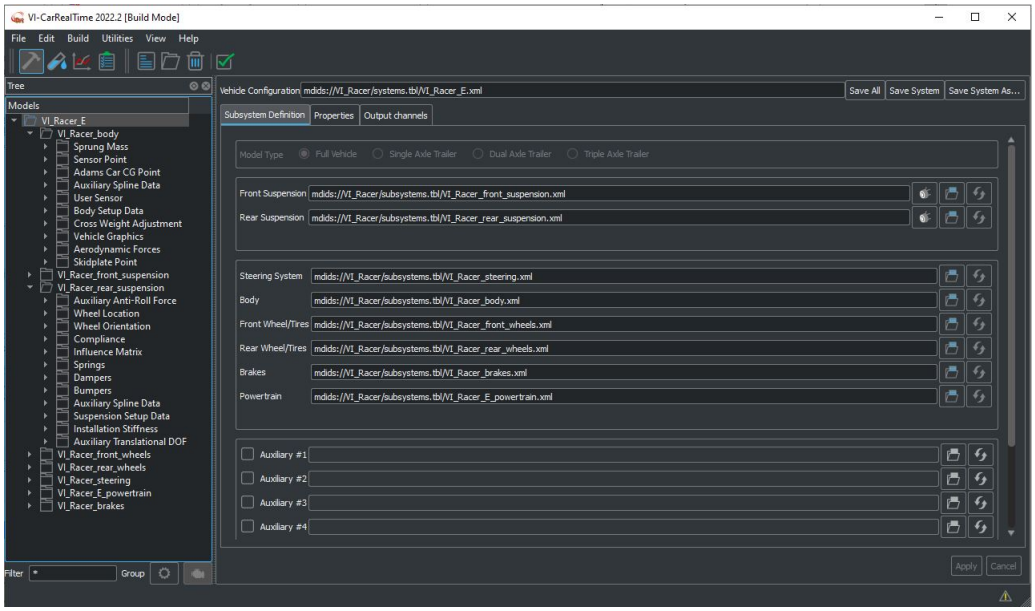


Figure 4.8: Interface of VI-CarRealTime 2022 to define the vehicle.

Defining all these aspects to make the simulated car behave similarly to the real vehicle is a challenging task. This thesis is not related into this aspect, but you can find more information in the thesis [8].

The interaction between VI-CarRealTime and Simulink enables the merging of VI-CarRealTime's accuracy in vehicle simulation with the controller design capabilities in Simulink. This facilitates testing and fine-tuning control algorithms in virtual environments, enhancing vehicle performance and safety through a MATLAB toolbox.

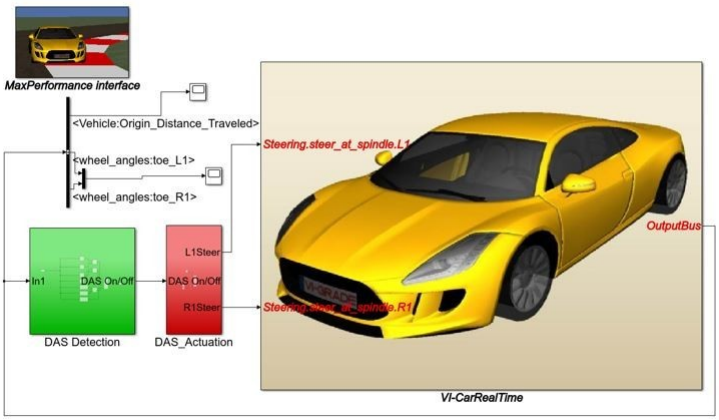


Figure 4.9: VI-Grade Simulink Interaction.

5 Yaw Rate Control Design

5.1 General Control Scheme

In this section, a general control scheme will be presented, taking into account both the inputs and outputs of this section, as well as the different elements that compose it.

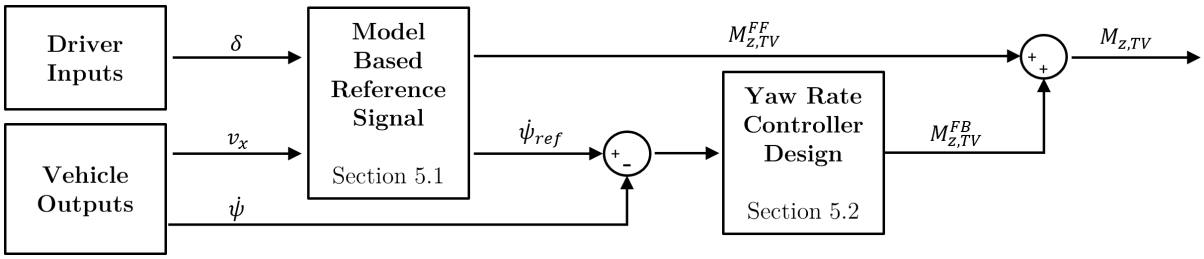


Figure 5.1: Yaw Rate Control Scheme.

The first element to be designed is the reference signal based on the model explained in Section 4.2, where we will study a reference behavior that can follow the CAT15x model. The second element to be designed is the Yaw Rate Controller, which will use the yaw rate reference obtained in Section 5.2 to, through a PID controller, meet that reference. The result of these two elements will be summed together to give way to the additional yaw moment reference that the Torque Allocator in Section 6 will have to fulfill.

It is important to mention that that this additional yaw moment is provided by the longitudinal forces of the electric motors, as the vehicle experiences other moments from forces such as lateral forces from the tires. This second yaw moment will not be taken into account since we cannot directly modify it as it is dependent on the vehicle's state.

5.2 Model Based Reference Signal

First, we need to define the desired reference behavior that we want our vehicle to have. For this purpose, the vehicle model discussed in Section 4.2 will be used as the current vehicle, and by modifying some of the model parameters, we will achieve the behavior of the reference vehicle [16].

Using these two models, the behavior of both will be compared in the same maneuver, in this case, Ramp Steer, the additional yaw moment needed for the current vehicle to have the behavior of the reference vehicle will be calculated as:

$$M_{z,TV}^{FF} = \left(\left(F_{y,f}^{ref} cos(\delta_w + F_{x,f}^{ref} sin(\delta_w)\right) l_f - F_{y,r}^{ref} l_r\right) - \left(\left(F_{y,f} cos(\delta_w + F_{x,f} sin(\delta_w)\right) l_f - F_{y,r} l_r\right) \tag{13}$$

where $F_{y,i}^{ref}$ $[N]$ and $F_{x,i}^{ref}$ $[N]$ are the lateral and longitudinal forces in the i axis from the reference model, $F_{y,i}$ $[N]$ and $F_{x,i}$ $[N]$ are the lateral and longitudinal forces in the i axis from the CAT15x model.

Therefore, the first step is to define this ideal behavior, which will be determined by comparing lateral acceleration vs wheel angle and slip angle.

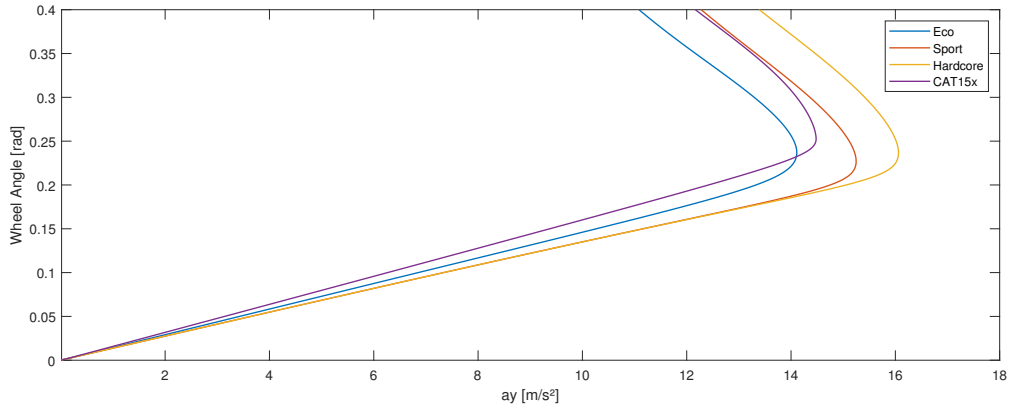


Figure 5.2: a_y vs δ for the differents modes.

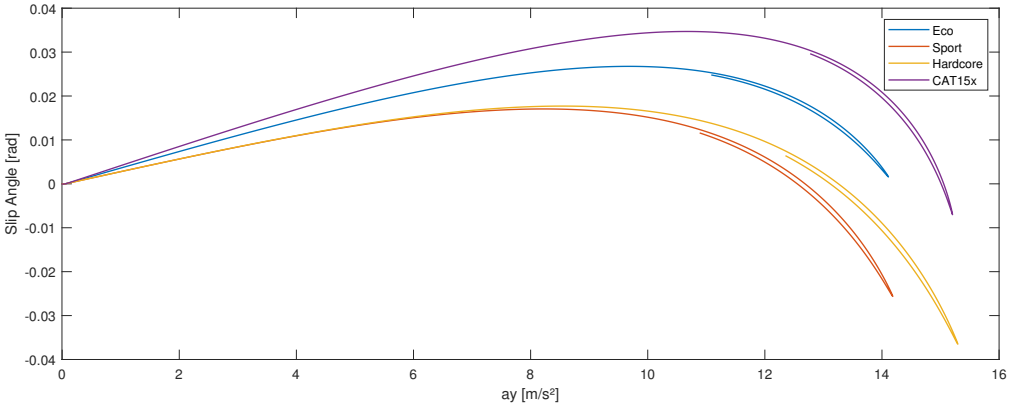


Figure 5.3: a_y vs β for the differents modes.

As observed in Figure 5.2 and 5.3, not only one behavior has been defined, but three possible behaviors with different objectives have been identified, referred to them as modes. The first reference signal or mode is called Eco and aims to make the vehicle more stable. The second mode is called Sport, with the goal of achieving higher yaw rate and lateral acceleration. This mode is considered the primary one, as in Formula Student, it is crucial to have a highly reactive car with high speed in curves and stability, considering the typical design of Formula Student racing circuits. Finally, the third mode is an exaggeration of the second, named Hardcore, in case an exceptionally aggressive level is required at any moment.

Once we have obtained the desired reference behavior, we will obtain both the additional yaw moment and yaw rate of the reference vehicle when the vehicle reaches its steady-state at different speeds and wheel angles. These two graphs, the additional moment and yaw rate reference, can be seen below in the 3D Figure 5.4. From this moment forward, all graphs, values, references, or results mentioned will be based using the Sport mode, as it is optimal for the general goal in Formula Student. Graphs for the other two modes can be found in the appendix A.

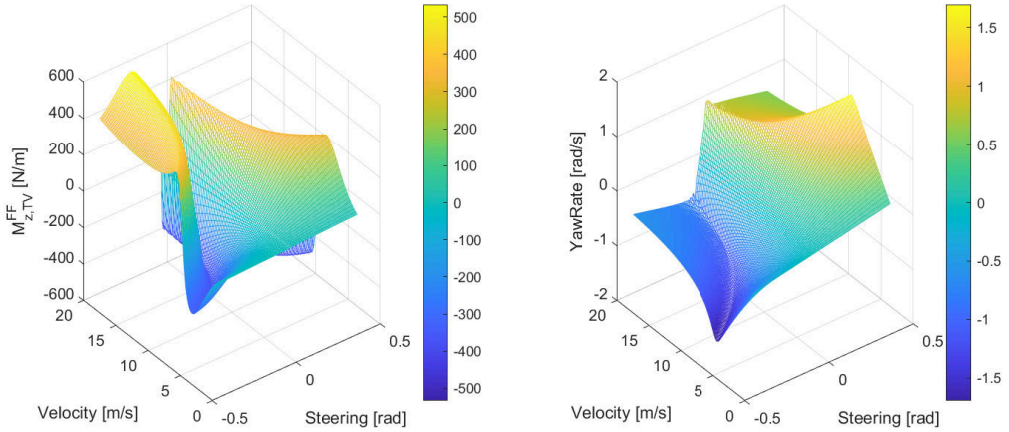


Figure 5.4: v_x vs δ vs $M^{FF}_{z,TV}$ and $\dot{\psi}_{ref}$ on Sport mode.

An important aspect to highlight is that, even though we now define three different behaviors, without being entirely sure if they are the best for the vehicle's performance or if the experimental car will reach the references, infinite data tables can be obtained by interpolating between two different modes. Even if the data for the additional moment and the reference signal are not directly related, it may be usefull experimentally finding an ideal behavior.

5.3 Yaw Rate Controller Design

PID control is a common used method in control engineering to stabilize dynamic systems. It adjusts the system output based on the proportion, integral and derivative of the difference between the reference and the current output, in this case, the yaw rate.

In addition, gain-scheduling [17] is a technique that adapts the controller parameters based on certain operating conditions. This method allows adjusting the controller's dynamics according to changes in working conditions, this leads to an improved system response in various situations.

To design this controller, it needs to be done in discrete time, considering the minimum communication frequency of the CAT15x. In this case, the Processing Unit (PU) can communicate at a high frequency and the bottleneck is found in the TPDO and the RPDO from the inverters.

The selected controller excludes the derivative component due to high-frequency noise in the input signal, which could be counterproductive. The notation for this controller is as follows:

$$C_z(v_x) = K_p(v_x) + K_i(v_x)\frac{T_s}{2}\frac{z+1}{z-1} \tag{14}$$

where K_p is the proportional constant value, K_i is the integral constant value, both in function of the velocity and Ts is the sampling time.

The two controller constants have been chosen using the Step Steer maneuver, performing the maneuver at various longitudinal speeds to identify the optimal constants. Some of the results obtained will be presented below, for different velocities and with a wheel steering step of $0,982\ rad$. Although most of the results can be found in the appendix A.

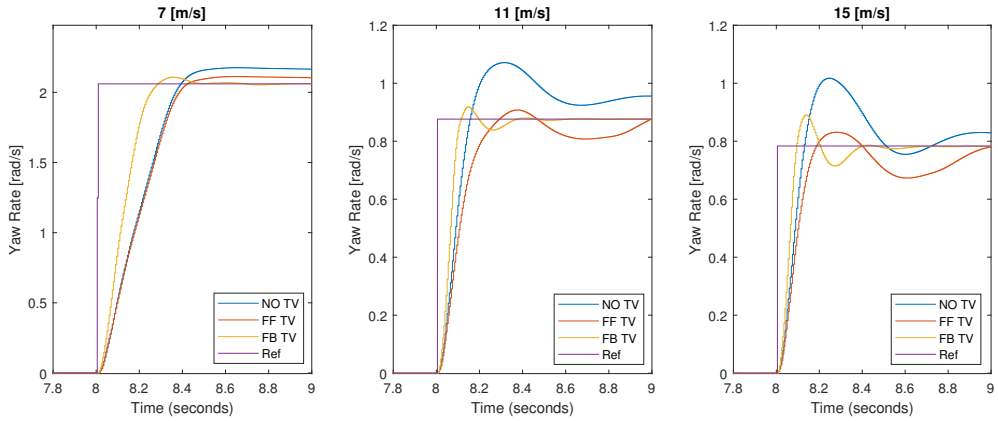


Figure 5.5: Step Steer response for different controllers and velocities.

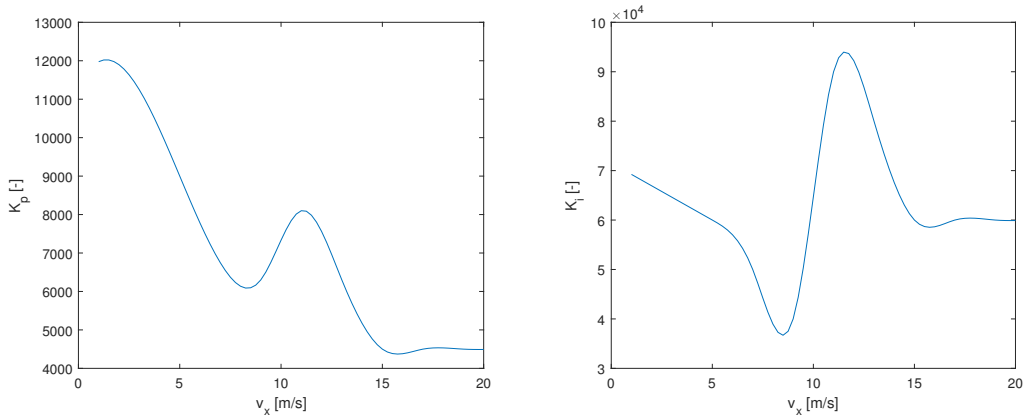


Figure 5.6: K_p and K_i plot.

5.3.1 Anti-Windup and Reset Component

In one hand, the Anti-Windup component is a mechanism designed to reduce errors related to the integrator. This component acts when the output is saturated, causing the controller to stop the integration. It introduces methods such as clamping, back-calculation or dynamic adjustments to the integral term, ensuring that the accumulated error does not become excessive during saturation periods. In this case, the method used is back-calculation.

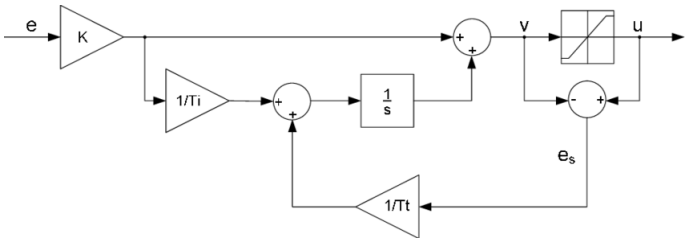


Figure 5.7: Back-calculation Antiwindup block diagram.

On the other hand, the reset component, often referred to as the integral term, addresses and eliminates the steady-state error. This integral action continuously adjusts the control output to bring the system closer to the desired setpoint, ensuring long-term stability. Reset value increases the controller's ability to eliminate persistent errors, but excessive values may lead to overshooting. In this controller the reset only depends on the steering and the velocity to avoid excessive integration.

6 Torque Allocator

Torque Allocator's main function is to distribute the available power of the vehicle among the different motors, in order to achieve the desired behavior from the Yaw Rate Controller, applying the additional yaw moment of its output. In the CAT15x case, having one motor per wheel means that this available power needs to be distributed among 4 motors. Having this motor distribution in a vehicle is the optimal scenario for implementing Torque Vectoring, as it allows for providing different power between front and rear as well as between left and right.

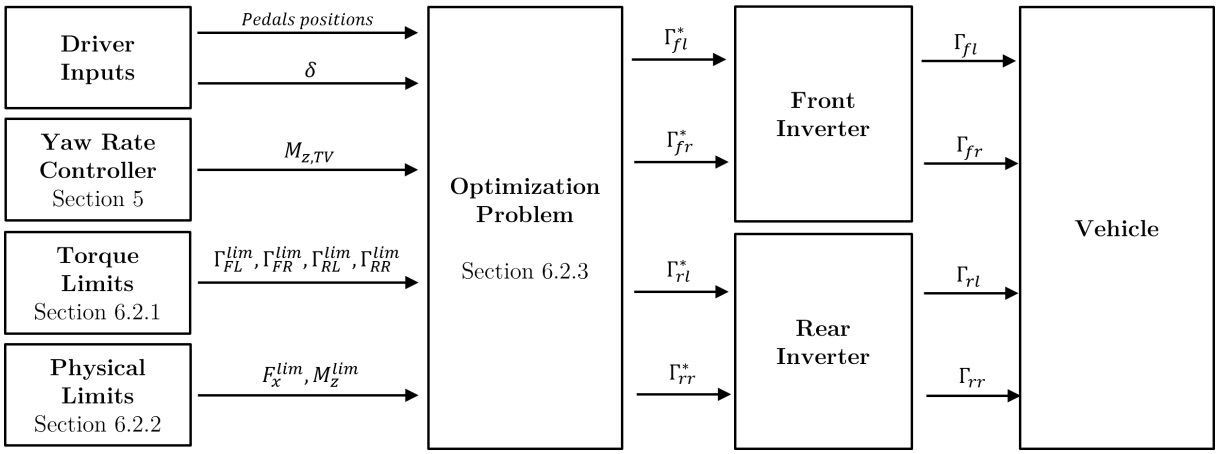


Figure 6.1: Torque Allocator Scheme.

6.1 Torque Distribution Control

First of all, it's necessary to define how the different inputs of the algorithm will be evaluated, such as F_x and M_z, which are defined by the throttle and the output of the Yaw Rate Controller respectively. To assess the different inputs, a 4WD vehicle dynamics model will be used, taking into account the wheel angle difference due to Ackerman steering geometry, as explained in Section 3.4.

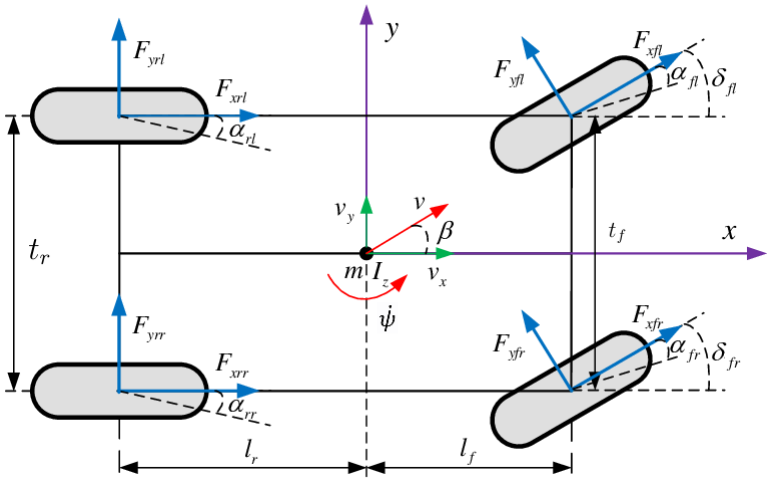


Figure 6.2: 4WD Vehcicle Dynamics Model.

$$F_x = \frac{\Gamma_{fl} \cdot cos\left(\delta_{w,fl}\right) + \Gamma_{fr} \cdot cos\left(\delta_{w,fr}\right) + \Gamma_{rl} + \Gamma_{rr}}{R_w} \tag{15}$$

$$M_z = \frac{\Gamma_{fl}\left(-cos\left(\delta_{w,fl}\right)\frac{t_f}{2} + sin\left(\delta_{w,fl}\right) l_f\right) + \Gamma_{fr}\left(sin\left(\delta_{w,fr}\right) l_r + cos\left(\delta_{w,fr}\frac{t_r}{2}\right)\right) - \Gamma_{rl}\frac{t_r}{2} + \Gamma_{rr}\frac{t_f}{2}}{R_w} \tag{16}$$

where $\Gamma_i\ [Nm]$ is the torque of each wheel, $\delta_{w,i}\ [rad]$ is the wheel angle, $R_w\ [m]$ is the wheel radius, $l_i\ [m]$ is the distance from the center of gravity of the vehicle to i track and $t_i\ [m]$ is the track width of the front of rear axis. In one hand, the variable $F_x\ [N]$ represents the longitudinal force desired by the driver through the throttle. On the other hand, the variable $M_z\ [Nm]$ stands for the additional moment.

6.2 Torque Distribution Optimization Problem

When thinking about solving the previous equations, we realize there are more variables than equations. Therefore, there are only two possible options: finding two more functions to make the equation system converge or opting for an optimization function, aiming to minimize the objective function.

So, a mix of both has been chosen by adding an additional equation explained later on. Since we still do not have a system of equations with the same number of variables as equations, using also an optimization method. While designing and implementing the algorithm in MATLAB, we chose to use an specific tool of this software. This function is called $mpcActiveSetSolver$, which solves quadratic programming (QP) problems and is part of the Model Predictive Control Toolbox.

This solver is useful for advanced MPC applications and custom QP applications, including applications that require C-code generation. This last specification is very important so all algorithms to be implemented in the PU is needed to be C-code generated for the proper functioning of the car and good communication between ROS.

$$J = \frac{1}{2}x^T Hx + f^T x \tag{17}$$

This is the structre of the objective function that solver uses, where x is the optimal solution, H is a Hessian matrix and f is the multiplier of the objective function linear term.

6.2.1 Torque Limits

In this problem since the output is the torques of each wheel, we need to give to the optimization problem their limits to be sure we never surpass some torque pipeline limitation. Those limitations comes from a lot of diferent sides, explained below, and need to select the smallest torque.

The first limitation comes from the Traction Control algorithm, TC output are the limits of the tire, both in acceleration and braking, using the Pacejka tire model explained in the Section 3.3, so the limitation is defined as:

$$\Gamma^{TC}_{min,i} \leq \Gamma_i \leq \Gamma^{TC}_{max,i} \quad \text{where} \quad i = fl, fr, rl, rr \tag{18}$$

The next limit comes from the Power Control algorithm, in Formula Student rules they make it clear that you cannot consume over 80kW battery power so PC ensures that this is fulfilled giving the total maximum torque taking into account all engines. This limitation is only for the positive torques because the regulation does not care about the regenerative power obtained.

$$\Gamma^{PC} \geq \sum_i \Gamma_i \quad \text{where} \quad i = fl, fr, rl, rr \tag{19}$$

The last limitation to ensure the performance and safety of the vehicle are two in one. The first element is to follow and not exceed the torque motor spline and the second one is to prevent overheating of inverters and motors. A LookUp Table was defined to work as a coefficient depending on the actual invterter and motor temperature for the torques given by the torque motor spline, which depends on the angular velocity of the wheel.

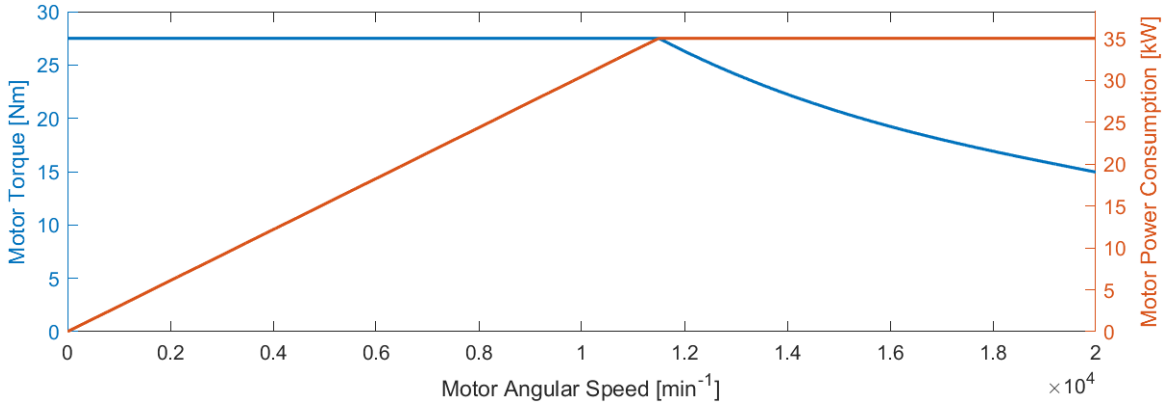


Figure 6.3: Torque motor spline. Angular speed vs Torque/Power consumption.

$$\begin{aligned} \Gamma^{spline}_{max,i} \cdot T_i = \Gamma^{temp}_{max,i} = -\Gamma^{temp}_{min,i} \quad \text{and} \\ \Gamma^{temp}_{min,i} \leq \Gamma_i \leq \Gamma^{temp}_{max,i} \quad \text{where} \quad i = fl, fr, rl, rr \end{aligned} \tag{20}$$

where T_i $[-]$ is the temperature coefficient.

Finally there is another limitation, but it is a user limitation, where you can change the maximum and minimum avaliable torque without needing to recompile, so we will not add it because it is not realted with torque vectoring algorithm. So after all this differents controls that limits torque we need to define an upper and lower boundary for each wheel.

$$\begin{aligned} \Gamma_{min,i} = min\left(\Gamma^{TC}_{min,i}, \Gamma^{temp}_{min,i}\right) \quad , \\ \Gamma_{max,i} = min\left(\Gamma^{TC}_{max,i}, \Gamma^{temp}_{max,i}\right) \quad \text{and} \\ \Gamma_{min,i} \leq \Gamma_i \leq \Gamma_{max,i} \quad \text{where} \quad i = fl, fr, rl, rr \end{aligned} \tag{21}$$

As the PC limitation is the sum of all the motor torques and not individually, we will add their limitation in another inequation on the optimization problem.

6.2.2 Physical Limits

Also, when a vehicle is turning the inner wheels do not have the same verical load as the outer wheels due to lateral load transfer. Now, using previous situation, as the torque limits change, also the maximum allowed longitudinal force and yaw moment changes so we need to know these limits to make sure the optimization problem will never make the car unstable.

This will be acomplished using Equation 15 and 16, using the torque limits given by the TC in this equations because we only want to restrict the physical limit of the car without

taking care about others boudaries. We will use this limits to use it as a restriction in the optimization problem.

$$F_{x,max} = \frac{\Gamma^{TC}_{max,fl} \cdot cos\left(\delta_{w,fl}\right) + \Gamma^{TC}_{max,fr} \cdot cos\left(\delta_{w,fr}\right) + \Gamma^{TC}_{max,rl} + \Gamma_{max,rr}}{R_w}^{TC}, \qquad (22)$$
$$F_{x,min} = 0$$

$$M_{z,max} = \frac{\Gamma^{TC}_{min,fl}\left(-cos\left(\delta_{w,fl}\right)\frac{t_f}{2} + sin\left(\delta_{w,fl}\right)l_f\right) + \Gamma^{TC}_{max,fr}\left(sin\left(\delta_{w,fr}\right)l_r - cos\left(\delta_{w,fr}\frac{t_r}{2}\right)\right)}{R_w}$$
$$\frac{-\Gamma^{TC}_{min,rl}\frac{t_r}{2} + \Gamma^{TC}_{max,rr}\frac{t_f}{2}}{R_w},$$
$$M_{z,min} = \frac{\Gamma^{TC}_{max,fl}\left(-cos\left(\delta_{w,fl}\right)\frac{t_f}{2} + sin\left(\delta_{w,fl}\right)l_f\right) + \Gamma^{TC}_{min,fr}\left(sin\left(\delta_{w,fr}\right)l_r - cos\left(\delta_{w,fr}\frac{t_r}{2}\right)\right)}{R_w}$$
$$\frac{-\Gamma^{TC}_{max,rl}\frac{t_r}{2} + \Gamma^{TC}_{min,rr}\frac{t_f}{2}}{R_w} \qquad (23)$$

6.2.3 Optimization Problem

Apart from this, one more input are taken into account and are the vertical loads of each tire. These are used to penalize the axis that have less vertical load and to reward the one that have more vertical load.

This will have two consequences, one direct and the other indirect. The direct consequence is that the axis with more vertical load will have more torque by forcing it in the optimization problem. The indrect consequence is that as the target of the optimization problem is to minimize the objective function, the solver will try to achieve the desired inputs with the minimum possible torque. The second one will help us to reduce the power consumption.

Finally the multi-objective problem will have the following structre:

$$min(J) \quad \text{where} \quad J = \sum_{n=1}^{3} \alpha_n \cdot J_n,$$
$$\text{subject to}$$
$$\Gamma_{min,i} \leq \Gamma_i \leq \Gamma_{max,i} \quad \text{where} \quad i = fl, fr, rl, rr, \qquad (24)$$
$$\sum_i \Gamma_i \leq \Gamma^{PC},$$
$$F_{x,min} \leq F_x \leq F_{x,max},$$
$$M_{z,min} \leq M_z \leq M_{z,max}$$

where J_n are the defined objectives and α_n $[-]$ are the priority coefficients for each of the TD objectives. The other nomencaltures has been explained in both 6.2.1 and 6.2.2. Now that the multi-objective has been stated, lets explain objective per objective.

The first objective refers to the longitudinal force, more specifically to the error between the desired longitudinal force by the driver and the longitudinal force given by the output of the optimization problem using the equation 15 and it is expressed as:

$$J_1 = ||F_x^{des} - F_x||^2 \tag{25}$$

where F_x^{des} $[N]$ is the desired longitudinal force demanded by the driver.

Also there is a priority coefficient to change the weight of this first objective depending on the state of the car. To know the state of the car the wheel angle has been used and to simplify the expressions it is considered absolute value. In one hand, when we observe this objective, we realize that the most important state to prioritize the longitudinal force is when the car goes in a stright line. On the other hand, in a close cornering, we do not want to priorize it because we will never use all the available power in four motors. Using this we can define this priority coefficient as follows:

$$\alpha_1 = \tau_1 \cdot (|\delta_w| + 1)^{\gamma_1} \tag{26}$$

where δ_w $[rad]$ is the wheel angle. This coefficient has the following shape.

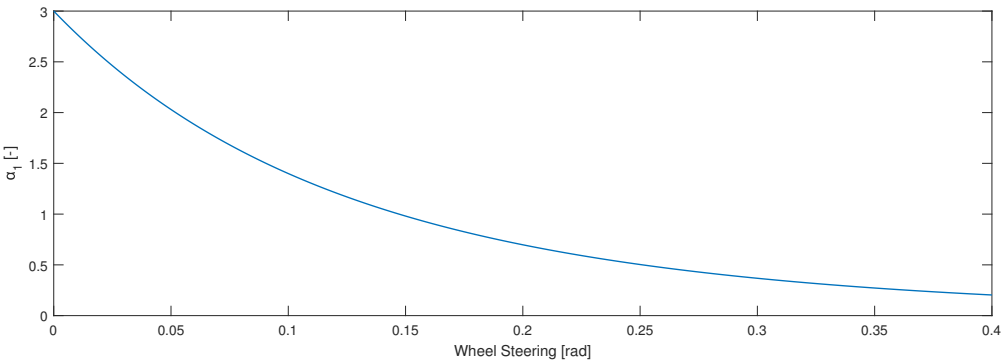


Figure 6.4: α_1 priority coefficient splie.

Continuing with the multi-objective problem, the second objective is refered to the error between the yaw moment desired given by the Yaw Rate Controller in Section 5 and the moment developed by the motors using the equation 16.

$$J_2 = ||M_z^{des} - M_z||^2 \tag{27}$$

As in the first objective, in this case we also want to vary the priority coefficient in function of the state of the car so we will use the same method but now we want to priorize this objective when the vehicle is in cornering and despise it in a stright line.

$$\alpha_2 = \tau_2 \cdot ln\left(\sqrt{|\delta_w|} + \gamma_2\right) \tag{28}$$

where δ_w $[rad]$ is the driver steering angle. This coefficient has the following shape.

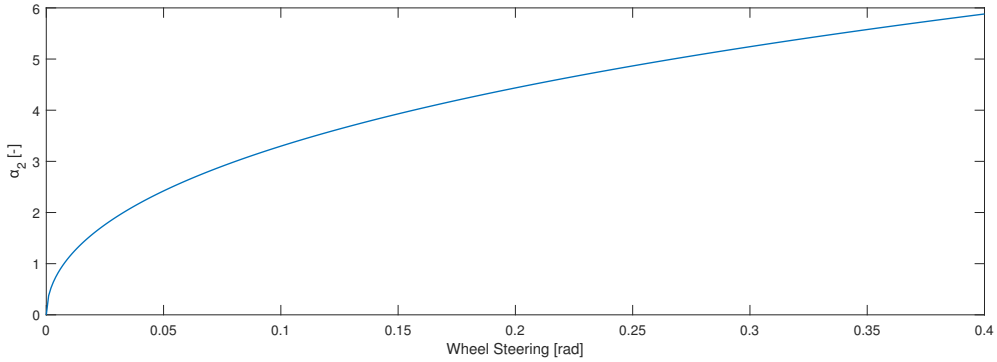


Figure 6.5: α_2 priority coefficient splie.

Different approaches have been taken, and it has been concluded that the best results are obtained when $\tau_1 = 3$ and $\gamma_1 = -8$ for the coefficient of prioritization of the longitudinal force and $\tau_2 = 12$ and $\gamma_2 = 1$ for the coefficient of prioritization of the additional yaw moment.

Finally, this last objective is a little bit different than the other ones, this objective is not an error and neaither an scalar, it is a 4x4 matrix where in the diagonal are the relative error of vertical loads from the other axle. Also, as this vertical loads change with the longitudinal state of the vehicle we do not need to change the priority coefficient depending on the wheel steering so it will be a constant value. The relative error can be expressed as:

$$\begin{aligned} \varepsilon_{z,f} &= \frac{\sum_j F_{z,fj}}{\sum_t \sum_j F_{z,tj}}, \\ \varepsilon_{z,r} &= \frac{\sum_j F_{z,rj}}{\sum_t \sum_j F_{z,tj}} \\ \text{where} \quad t = f, r \quad &\text{and} \quad j = l, r \end{aligned} \tag{29}$$

where $F_{z,tj}$ $[N]$ are the different vertical loads for each wheel, t defines if the load is from the rear or the front and j if it is from the left or the right.

And the third objective can be defined as:

$$J_3 = \begin{bmatrix} \varepsilon_{z,r} & 0 & 0 & 0 \\ 0 & \varepsilon_{z,r} & 0 & 0 \\ 0 & 0 & \varepsilon_{z,f} & 0 \\ 0 & 0 & 0 & \varepsilon_{z,f} \end{bmatrix} \tag{30}$$

Also, to determine if the optimization problem is minimizing the objective function and the result is converging, there is a variable, already integrated into the MATLAB solver used, called $exitflag$. This variable can take the following values:

- > 0: The solution is optimal. In this case, exitflag represents the number of iterations performed during optimization.
- $= 0$: T he maximum number of iterations was reached. The solution might be suboptimal or infeasible.
- $= -1$: The problem appears to be infeasible, that is, the constraint $Ax \leq b$ cannot be satisfied.
- $= -2$: An unrecoverable numerical error occurred.

7 Torque Vectoring Results

Once all the necessary theoretical and technical aspects have been discussed for designing and developing a Torque Vectoring in a Formula Student vehicle and to distribute power across all four motors, the next step is to showcase the various results of all control system modules to verify their functionality and performance.

These mentioned modules include the Yaw Rate Controller in Section 5, discussing the model in which the controller was based in Section 4.2, and the design of the closed-loop controller detailed in subsection 5.3. Another crucial module is the optimization problem of torque distribution in Section 6.

Due to various testing limitations and constraints, these results will be compared under different conditions involving various components of the Yaw Rate Controller, both Feed Forward or Feed Back component. Table 2 outlines all the different combinations and maneuvers, indicating which data is available, providing crucial insight for data analysis.

Also, to analyze the data and the vehicle's behavior, we will focus on representing a similar driving scenario for each type of controller used. This will be done through the simulation tool employed, which performs a series of iterations to find an optimal trajectory path and a speed profile for the "free" vehicle with the controller disabled.

These iterations cannot be performed with the other two types of controllers due to the unavailability of licenses for this tool. Therefore, the same trajectory and speed profile will be used for all controller types. It is important to mention that this does not limit the vehicle's behavior. By using the simulation tool in this way, we employ the same path and speeds, but in none of the three cases (TV OFF, TV FF, TV FF+FB) will the driver use the same inputs for the steering wheel and pedals. This approach not only allows us to analyze the improvement in the vehicle's behavior but also demonstrates the potential for improvement in the vehicle's performance with the controller.

	Controller Type	**Ramp Steer**	**Skid Pad**	**AutoX**
Real Data	**OFF**	✓	✓	✓
Simulation Data	**OFF**	✓	✓	✓
	FF	✓	✓	✓
	FF + FB	✓	✓	✓

Table 2: Table displaying the collected data from the various maneuvers.

To perform a proper analysis of all the aspects required for the effective operation of Torque Vectoring, various factors will be examined in all the maneuvers and circuits mentioned in the above table. We will compare different aspects by observing the vehicle's behavior with Torque Vectoring turned off (TV OFF) and turned on (TV ON). The aspects that will be analyzed include:

- Vehicle Behaviour
  - Reference $\dot{\psi}$ vs Actual $\dot{\psi}$
  - Slip Angle β
  - Lateral Acceleration a_y
  - Peak values
- Torque Distribution Problem
  - Reference F_x vs Actual F_x
  - Reference M_z vs Actual M_z
  - Priority coefficients and $exitflag$.
  - Torques

In the Vehicle Behavior section, the analysis will also take into account the performance of Torque Vectoring only with the Feed-Forward component enabled.

7.1 Vehicle Behaviour

7.1.1 Ramp Steer

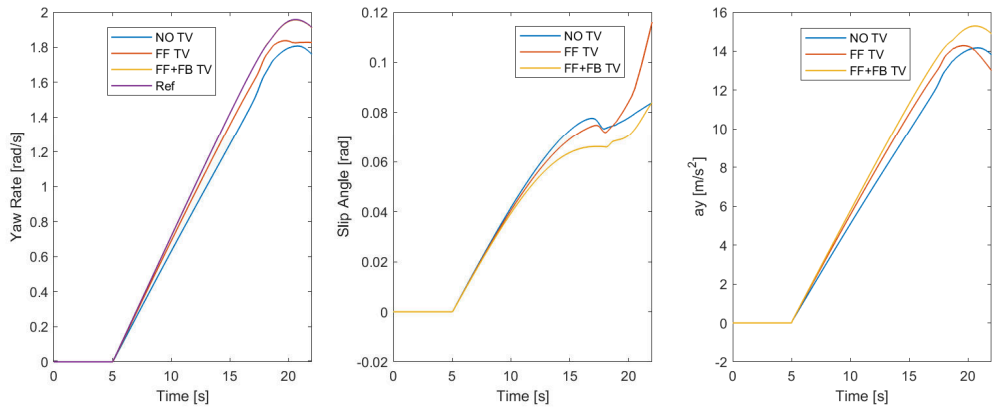


Figure 7.1: Ramp Steer Vehicle Behaviour Results.

In Figure 7.1, it is noticeable how the model-based Feed-Forward component mentioned in Section 5.2 operates. However, it is not enough to achieve the yaw rate reference, and with the PID controller, it is able to accomplish this task. Additionally, as expected, the vehicle gets higher lateral acceleration and lower slip angle, reaching its maximum potential with feedback.

Details	Nomenc.	Unit	NO TV	FF TV	FF+FB TV
Peak Yaw Rate	$\dot{\psi}$	$[rad/s]$	1,807	1,8377	1,956
Peak Slip Angle	β	$[rad]$	0,0838	0,1161	0,0839
Peak Lateral Acc.	a_y	$[m/s^2]$	14,1711	14,2864	15,2929

Table 3: Ramp Steer Peak Values.

7.1.2 Skid Pad

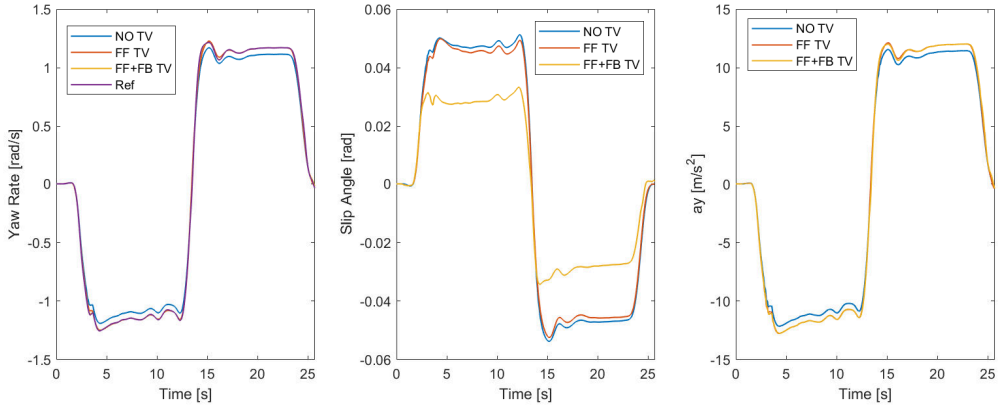


Figure 7.2: Skid Pad Vehicle Behaviour Results.

Going to the Skid Pad track, as can be seen in Figure 7.2, the increase of the yaw rate or the lateral acceleration is not as big as it could be. Nevertheless, the significant improvement lies in the reduction of the slip angle, providing an opportunity for potential enhancements in order to maximize the vehicle's performance.

Details	Nomenc.	Unit	NO TV	FF TV	FF+FB TV
Peak Yaw Rate	$\dot{\psi}$	$[rad/s]$	1,171	1,227	1,215
Peak Slip Angle	β	$[rad]$	0,0513	0,0499	0,0333
Peak Lateral Acc.	a_y	$[m/s^2]$	12,150	12,759	12,756

Table 4: Skid Pad Peak Values.

7.1.3 AutoX

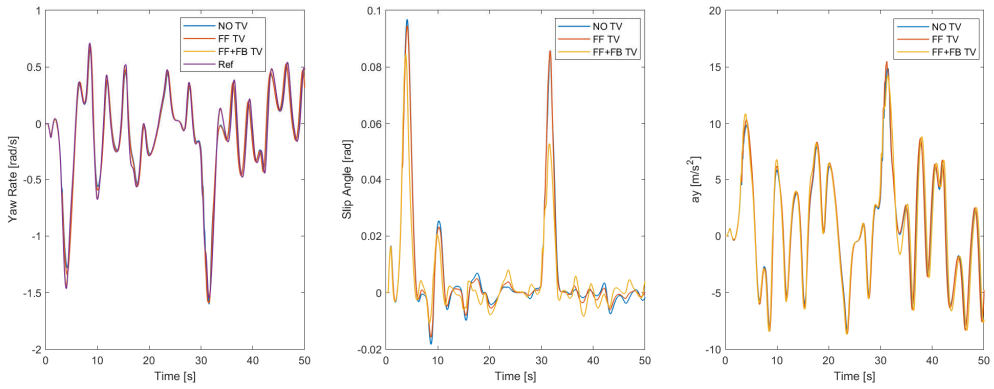


Figure 7.3: AutoX Vehicle Behaviour Results.

Finally, on the AutoX track in Figure 7.3, if we examine the peak values in all three graphs, it can be observed that in the first one, the highest yaw rate values are achieved when both FF and FB components are enabled. In the second and third graphs, during these peaks that push the vehicle to its limits, the controller stabilizes it, resulting in lower slip angles and lateral accelerations. However, this does not occur throughout the entire run. In those track segments where the vehicle does not reach its limits, the slip angle and lateral acceleration using the TV controller are higher.

Details	Nomenc.	Unit	NO TV	FF TV	FF+FB TV
Peak Yaw Rate	$\dot{\psi}$	$[rad/s]$	0,737	0,773	0,781
Peak Slip Angle	β	$[rad]$	0,0968	0,0946	0,0846
Peak Lateral Acc.	a_y	$[m/s^2]$	14,955	15,510	14,243

Table 5: AutoX Peak Values.

7.2 Yaw Rate Controller

As demonstrated in the Figures 7.1, 7.2 and 7.3, the results looking to the behaviour of the car are really impresive. However, it is not only important the behaviour of the car, the correct functioning of all different elements of the controller also has to be checked and the first one is the Yaw Rate Controller, designed in Section 5.

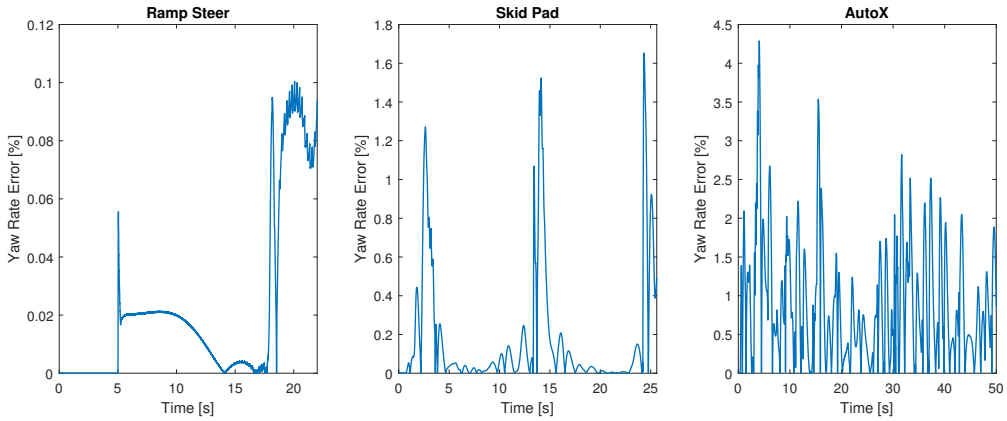


Figure 7.4: Error between actual and reference yaw rate.

In the Figure 7.4 above, the relative error calculated using the equation 12, between the yaw rate reference and the actual yaw rate of the vehicle using the Yaw Rate Controller are observed for each different maneuver studied. Looking to these errors it can be stated that the controller works properly as the average error of all three different conditions are:

	Unit	**Ramp Steer**	**Skid Pad**	**AutoX**
Average error	[%]	0,0218	0,1797	0,8872

Table 6: Average error between actual and reference yaw rate.

7.3 Torque Distribution Problem

In the following sections, the output of the TD Problem, as explained in Section 7, will be analyzed. The comparison between the current data and the reference data for both the longitudinal force and the additional yaw moment can be found below. This comparison is conducted for each situation, as mentioned in the Vehicle Behaviour Results in Section 7.1. Additionally, a plot showing the result of the optimization problem for all four different torques is provided.

At the end of this section, there will be a review to discuss the results obtained. Apart from this, not within this section but in the apendix B, you can find a graph of the priority coefficients and the exitflag value varying with time for the optimization problem in the three different cases.

7.3.1 Ramp Steer

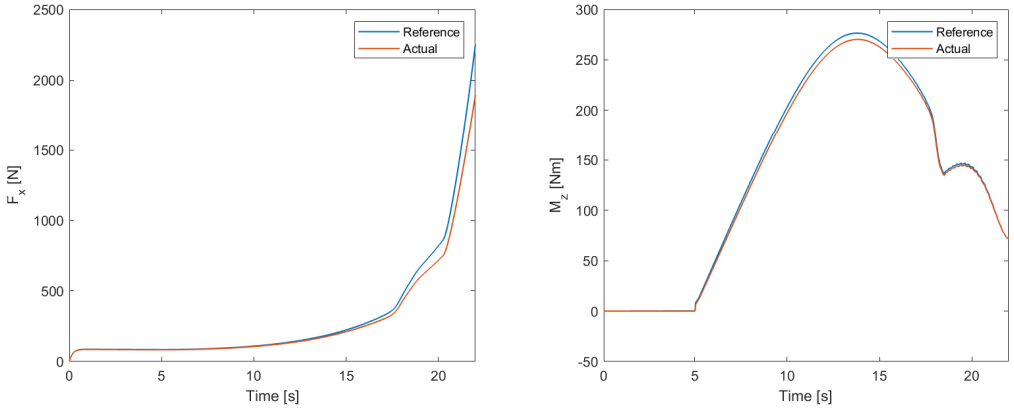


Figure 7.5: Ramp Steer Torque Distribution Objectives Results.

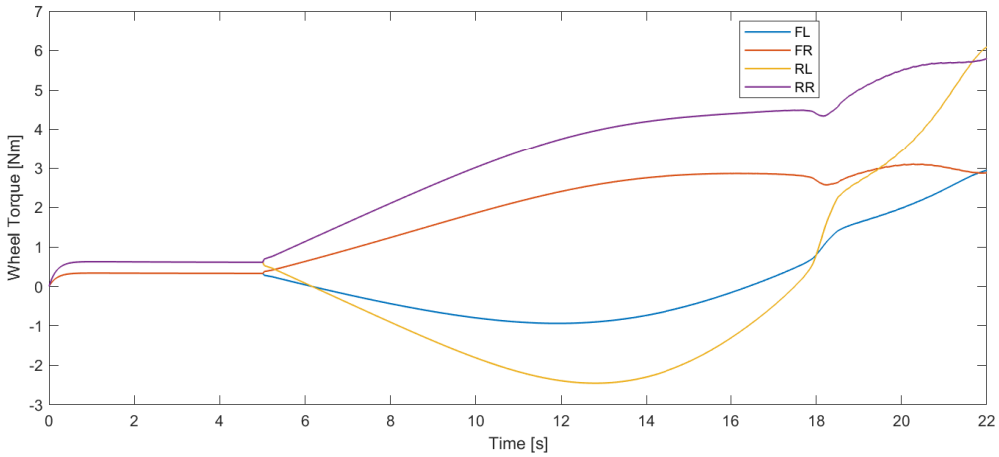


Figure 7.6: Ramp Steer Torques Results.

7.3.2 Skid Pad

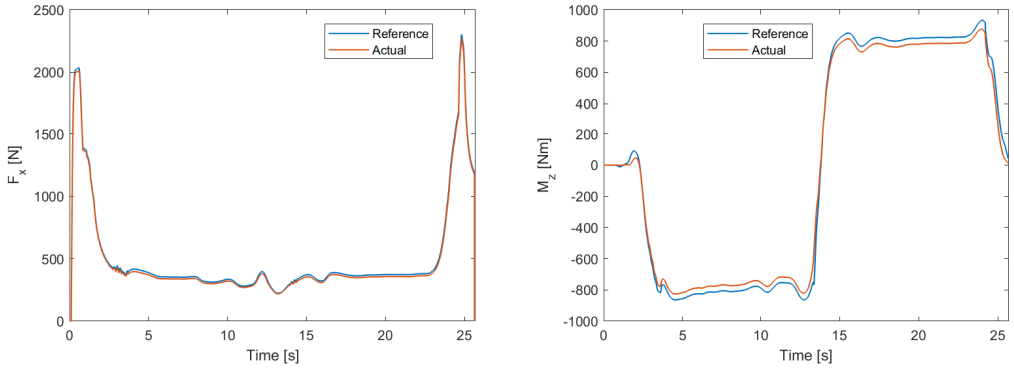


Figure 7.7: Skid Pad Torque Distribution Objectives Results.

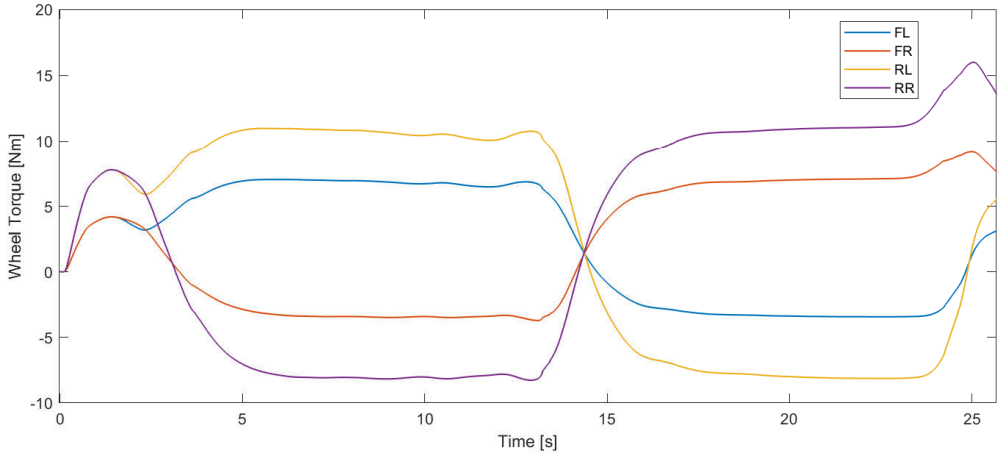


Figure 7.8: Skid Pad Torques Results.

7.3.3 AutoX

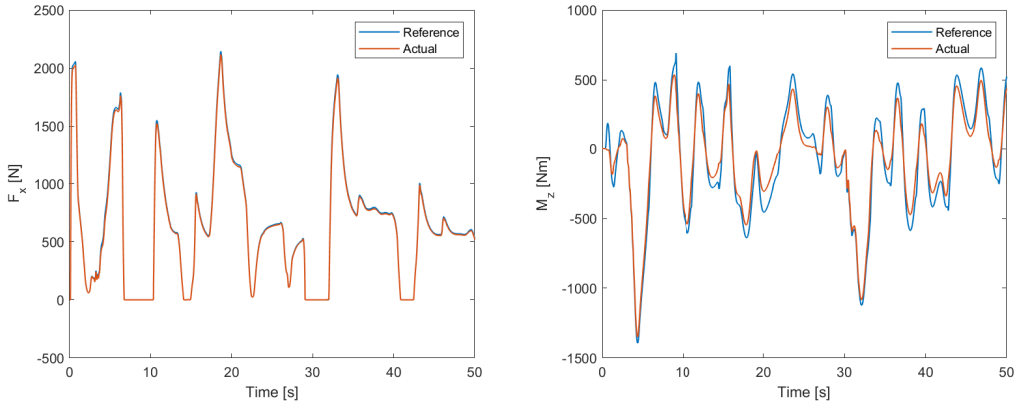


Figure 7.9: SkidPad Torque Distribution Objectives Results.

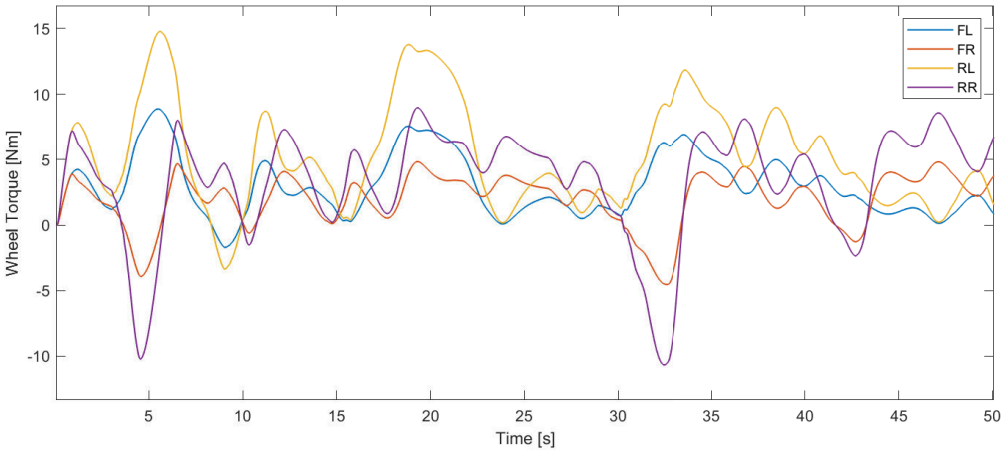


Figure 7.10: AutoX Torques Results.

7.3.4 Review

Analyzing the figures, we can conclude that the operation of the Torque Distribution Problem works correctly, meeting its respective objectives with minimal error in all situations. If we look at the output of the TD Problem, we can observe that it has a direct relationship with the reference signals, without inconclusive results.

	Unit	**Ramp Steer**	**Skid Pad**	**AutoX**
F_x average error	[%]	2,168	0,378	0,325
M_z average error	[%]	1,161	4,262	5,259

Table 7: Average error between actual and reference function objectives.

In Table 7, the average error for both input reference signals can be observed. Considering this as a relatively small error assures the correct following of references. Additionally, not in this section but in the appendix B, you can find a graph showing the relative error between the current signal and the reference signal for both the longitudinal force and the yaw moment, along with its average value, similar to what is presented in Section 7.2.

8 Economical Analysis

This economical analysis considers personal costs, divided into various stages, material costs and the costs of the software used. The budget does not include the complete cost of the vehicle and its manufacturing. Most of the different parts, both physical and software, have been designed for students at no cost and are manufactured for free by external companies or for a cheaper price. Therefore, only the costs of using both software and hardware directly involved in the Torque Vectoring algorithm are taken into account.

8.1 Personal Cost

The personal cost is divided in the diferents tasks and steps in which the development of Torque Vectoring has involved. This tasks, called details, goes from the start of the project to the instance before present this thesis.

Details	**Time [h]**	**Cost [€/h]**	**Total Cost [€]**
Previous Research	100	20	2.000
Model Design	60	20	1.200
Model Validation	35	20	700
Yaw Rate Controller Design	40	20	800
Torque Allocator Design	70	20	1.400
Simulink Implementation	120	20	2.400
Simulation Validation	80	20	1.600
Test Preparation	10	20	400
Real Test Validation	60	20	1.200
Data Analysis	60	20	1.200
TOTAL	**635**		**12.700**

Table 8: Personal cost chart.

8.2 Material Cost

For the material cost of the project, only the essential material will be adressed, since the budget for the powertrain like electric mototrsm battery or inverters are not a matter to discuss in this work. So, the budget will adress the necessary harware for design and implement the Torque Vectoring.

Details	Units	Cost [€/unit]	Total Cost [€]	Real Cost [€]
Computer	1	800	800	800
Processing Unit	1	800	800	800
MLP-25	4	777,05	3.108,2	3.108,2
Vectornav VN-300	1	4.850	4.850	0
TOTAL			**9.558,2**	**4.708,2**

Table 9: Material cost chart.

8.3 Software Cost

Finally, an important part or more precisely the most important part are the software used and with it the license to use it freely. In this section only both MATLAB and VI-Grade are taked into account.

Details	Cost [€/year]	Real Cost [€/year]
MATLAB License	10.000	0
VI-Grade License	10.000	0
TOTAL	**20.000**	**0**

Table 10: Software cost chart.

8.4 Total Cost

Details	Amount [€]
Personal Cost	12.700
Material Cost	4.708,2
Software Cost	0
TOTAL	**17.408,2**

Table 11: Total cost chart.

9 Enviromental Impact

Car electrification has emerged as a crucial response to the environmental and energy challenges we currently face. We can see their relevance in reducing carbon emissions and dependence on fossil fuels. These cars offer a cleaner and more sustainable alternative in front of iternal combustion vehicles, driving the transition towards more eco-friendly mobility.

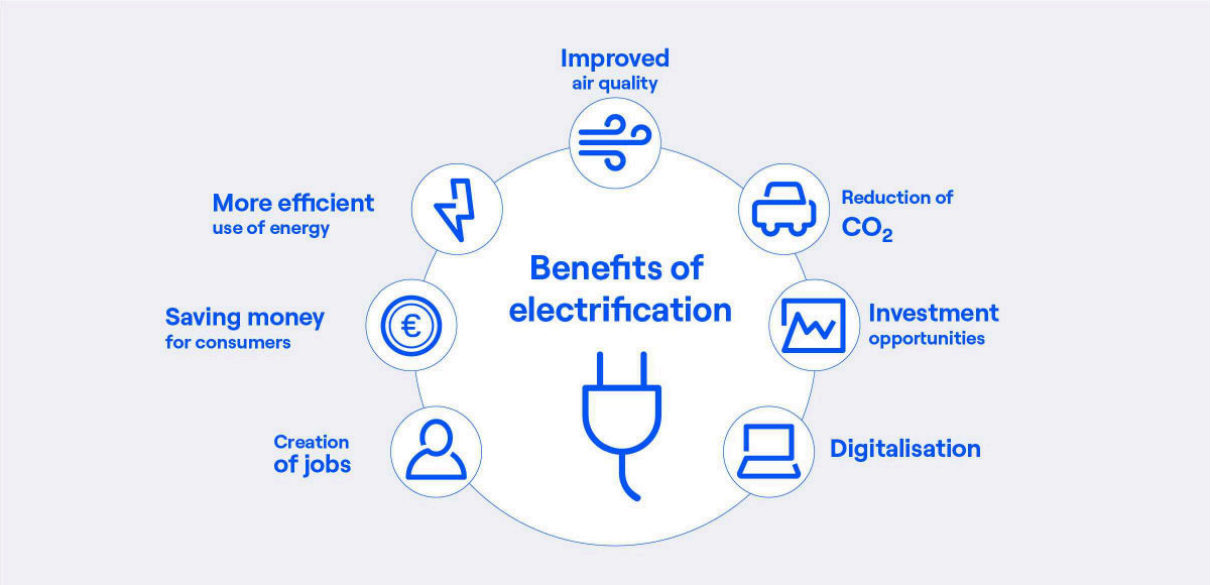


Figure 9.1: Benefits scheme of electrification.

In this context, the Formula Student competition has played a fundamental role. In its EV class, it has been an extraordinary testing ground for the development of innovative technologies seeking efficiency, performance, and sustainability in high-performance electric vehicles. Additionally, it serves to raise awareness and educate future engineers about electric mobility and its pros and cons.

Torque Vectoring, one of these technologies and the focus of this paper, has been a true revolution in the design of cars, with a bigger impact in the competitions ones. It allows independent control of the power delivered to each wheel, enhancing traction, stability, and maneuverability. Thanks to the benefits offered by Torque Vectoring, positive aspects related to its environmental impact can be observed. One remarkable impact is that it contributes to decreasing tire wear, which is costly in environmental and energy terms. This precise control over all four wheels aids in minimizing these environmental and energy costs.

10 Social and gender equality assessment

This thesis has been completed thanks to the Formula Student competition. In my case, BCNeMotorsport is sponsored by ETSEIB and ETSETB. Currently, in these engineering fields, most of the students are men, and the same trend applies in motorsport. Hence, it is predictable that the team will have more men than women, which is observable in most of the Formula Student teams. Over the years, the participation of women in both fields has increased significantly. This rise is due to the actions taken by engineering degrees and motorsport to improve women's participation as teams like Ferrari or McLaren have specific motorsport programs for women.

Despite this issue, the roles women have taken in the team demonstrate that there is no discrimination based on gender. Additionally, nationality, social status or beliefs do not matter. The only mandatory requirement is enrollment in one of the two universities, as shown in the different alumnis, but some costs to stay in the team like some testing and competitions meals are carried by the students.

Another factor to consider is that in the Formula Student project, a minimum level of knowledge is required due to the complexity of various scopes. Joining a team requires a CV, as well as a motivation letter, which carries more weight in the decision-making process. The majority of the tasks are not directly related to the degree or master's program so the main skill required is interest and motivation. There are numerous studies and mechanisms available to verify whether the participation of women in engineering degrees and motorsport has increased.

Throughout this project, careful attention has been paid to gender aspects by using non-sexist or androcentric images and employing inclusive language, such as using each person's full given name.



Figure 10.1: Gender equality, diversity and social inclusion diagram.

Conclusions

To conclude this project, the first step is to discuss the different objectives and to execute an analysis on each of them. The main objectives can be divided into four. The first one is to define the entire theory of vehicle dynamics 3 used both in this project and in many others. Besides, it can serve as a foundation and be of great help for future generations of BCNeMotorsport's Vehicle Control department members. In this first objective, we will also add defining a model-oriented control 5.2, where the model's results 4.2.1 have been very good, still allowing margin for improvement in future years.

The second main objective is the design and implementation of a yaw rate control 5.3, generating a reference using the employed model and adjusting to that reference through two components, the Feed-Forward component and the Feed-Back component. It is important to highlight the substantial enhancement provided by the PID controller results after applying gain-scheduling 5.6 and the versatility of the different proposed modes 5.4.

Regarding the torque distribution, which is the third goal, it is evident that the multi-objective optimization function 6.2 has generated outstanding results, closely following the objective function with minimal computational cost and a very fast response.

Finally, the last main objective are the results. The overall results of the controller clearly demonstrate its effective functioning and the good communication and interaction among the different elements of the algorithm, as well as the improvement in the vehicle's behavior 7.1. Additionally, the VI-CarRealTime simulation tool and its implementation in Simulink have been learned and used perfectly, which was not an easy task. The only downside, attributable to external factors unrelated to this project, has been the inability to complete the implementation and analyze the results on the actual prototype.

Future Work

Regarding to the future work to follow up with this thesis, there are several paths to enhance vehicle behavior and performance. One first improvement that can be applied is to use a more precise model as an orientation reference for yaw rate. Also, with the yaw moment obtained from this new model, the experimental vehicle can better align with the reference without needed of the PID controller, making the control more accuracy.

Nevertheless, without a doubt, the main path to continue with the project is to conduct the same tests performed in simulation with the experimental vehicle. Although the simulation model behaves very similarly to the CAT15x, it does not take into account the communication frequency of actuators and sensors, their data noise, and the variation between different test locations.

Additionally, the Model Based Reference Signal 5.2 will need an adaptation, either by using interpolation methods or by analyzing data during the test and designing a new behavior for the vehicle. In this way, the driver can also provide valuable feedback on the vehicle's performance. Additionally, scaling the PID coefficients to work well experimentally will be necessary to complete the process.

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